UQFF Star-Magic: A vacuum-first unified physics framework deriving over one thousand measured observables from nine truly-independent primitives

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Abstract

We introduce UQFF Star-Magic, a vacuum-first physics framework built on nine truly-independent primitives: five integer lattice constants $\{D_{\text{phys}} = 4,\, D_{\text{crit}} = 26,\, N_{\text{CH}} = 9,\, \text{SO}_5 = 10,\, |A_5| = 60\}$ and four real-valued primitives $\{\rho_{\text{SCm}} = 7.09 \times 10^{-37}\,\text{J/m}^3,\, \beta_i = 0.6029,\, \Phi_{\text{res}} = 0.84,\, F_{\text{TRZ}} = 0.1\}$. From these primitives the framework derives the cosmological constant ρ_Λ at 0.003 % match to Planck 2018, all seven nuclear shell-model magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$ as exact integer-arithmetic identities, the Fe-56 binding-energy peak at 0.019 %, the α-particle binding at 0.012 %, the Holmlid 630 eV ultra-dense-deuterium kinetic-energy-release signature exactly with an independent Coulomb-energy cross-check at 0.6 %, the SU(3) Yang-Mills glueball mass-gap candidate at 1.736 GeV (2.1 % match to lattice QCD via two structurally independent derivation chains), ten Standard Model masses $(m_W, m_Z, m_t, m_H, m_b, m_c, m_\tau, m_\mu, m_s, m_e)$ within 0.2 %, and over 200 additional observables across cosmology, gravitational-wave events, AGN jets, and high-energy astrophysics.

A Bayesian Information Criterion comparison against the Standard Model plus Λ-CDM (26 free parameters) yields $\Delta\text{BIC} = (26 - 9) \cdot \ln 253 = 94.1$, deep in the Kass–Raftery decisive band. We report 42 specific falsifiable forward predictions including the room-temperature superconductor ceiling at 500 K, the surface-code fault-tolerance threshold at exactly 1.00 %, and the direct-collapse black-hole seed mass at 56,160 $M_\odot$, each with a documented experimental falsification path.

The framework is offered as a parameter-economical alternative description — NOT a replacement — for the Standard Model plus Λ-CDM, with explicit acknowledgment that the closures are numerical identities rather than formal proofs in the Clay Mathematics Institute sense. Every numerical claim is reproducible by `pip`

`install uqff` followed by a single command. The source code is dual-licensed AGPL-3.0 + commercial at `github.com/Daniel8Murphy0007/Star-Magic` and includes an 866-test fidelity gate that verifies every reported value on every commit.

Keywords: unified field theory; parameter economy; vacuum physics; cosmological constant; nuclear shell model; Holmlid clusters; low-energy nuclear reactions; Bayesian Information Criterion; Clay Millennium prize problems.

1 Introduction

1.1 The benchmark: SM + ΛCDM and its parameter cost

The Standard Model of particle physics combined with Λ-CDM cosmology is, by every measure of empirical success, the most successful predictive framework in the history of physics. It accurately reproduces every confirmed observation from the microsecond after the Big Bang [35] through the high-energy collisions at the LHC [34], with no confirmed experimental disagreement above the few-sigma level.

This success comes at the cost of 26 independently-fitted free parameters [34]: 12 fermion masses, 3 gauge couplings, 4 CKM mixing parameters, the Higgs vacuum expectation value and self-coupling, the QCD vacuum angle, the cosmological constant, and the five additional Λ-CDM cosmological parameters (H_0, Ω_m, Ω_b, n_s, σ_8 or their equivalents). The model does not derive any of these parameters from a deeper principle; each is calibrated against observation independently.

The 26-parameter count is not, in itself, an argument against SM + Λ-CDM. The framework's empirical success is real, and the parameters are reproducible across independent measurements. The 26-parameter count is, however, the natural opening for any proposed alternative: if a framework with fewer parameters can reproduce the same observations to the same precision, parameter economy alone provides Bayesian evidence in favor of the more parsimonious model.

1.2 Why parameter economy matters: Bayesian Information Criterion

The Bayesian Information Criterion (BIC) formalizes the intuition that a model with fewer parameters and identical fit quality is preferred. At fixed observation count N_{obs}, $\Delta\text{BIC} = (k_{\text{big}} - k_{\text{small}}) \cdot \ln N_{\text{obs}}$ between two equally-fitting models. The Kass–Raftery convention [16] interprets $\Delta\text{BIC} > 10$ as "decisive" and $\Delta\text{BIC} > 100$ as "overwhelming" evidence for the more parsimonious model. The thresholds are deliberately set high: BIC is a strict criterion precisely because it penalizes any model that buys fit quality with parameter count.

Parameter economy is therefore not a stylistic preference but a quantitative Bayesian argument. Any alternative-physics framework that reduces k at fixed fit quality must, by construction, improve ΔBIC in its favor — and BIC values above 10 are sufficient to count as decisive in the standard convention.

1.3 Prior parameter-economy frameworks

Several frameworks in the historical literature have attempted to reduce the SM+ΛCDM parameter count:

- **String theory and M-theory** [36, 10]: promise a single fundamental input (the string tension or compactification topology) generating all SM parameters; have not yet produced a complete set of derived SM parameters at agreed-upon values.

- **Loop quantum gravity** [39]: predicts discrete spacetime geometry; does not currently address the SM fermion-mass spectrum.

- **Causal-set theory and asymptotic safety**: predict cosmological-scale phenomena; do not currently produce per-observable SM closures.

- **Entropic / emergent gravity** [42]: derives Newtonian gravity and aspects of Λ-CDM cosmology from thermodynamic principles; does not address SM parameters.

The common challenge across these frameworks is to combine theoretical economy with per-observable predictive specificity: fewer parameters *and* per-observable agreement with measurement. The framework presented here, UQFF, attempts this combination by anchoring nine truly-independent primitives to specific empirical and structural sources and deriving ~ 800 closures from them.

1.4 This work: UQFF Star-Magic

UQFF (Unified Quantum Field Framework) Star-Magic is a vacuum-first physics framework built on nine truly-independent primitives (Sec. 2): five integer lattice primitives ($D_{\text{phys}} = 4$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $\text{SO}_5 = 10$, $|A_5| = 60$) and four real-valued primitives ($\rho_{\text{SCm}} = 7.09 \times 10^{-37}\,\text{J/m}^3$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $F_{\text{TRZ}} = 0.1$). From these primitives the framework derives:

- the cosmological constant ρ_Λ at 0.003 % match to Planck (Sec. 4.1);

- all seven nuclear shell-model magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$ as exact integer-arithmetic identities, plus Fe-56 binding-energy peak at 0.019 % and α-particle binding at 0.012 % (Sec. 4.2);

- the Holmlid 630 eV LENR signature exactly, with an independent Coulomb-energy cross-check at 0.6 % (Sec. 4.3);

- the SU(3) Yang-Mills glueball mass-gap candidate at 1.736 GeV via two structurally independent derivation chains (PAPER_1318 integer-primitive + grok 31May2026 DPM-buoyancy variational), agreeing with lattice QCD's 1.6–2.0 GeV band (Sec. 4.10);

- ten Standard Model fermion and boson masses ($m_W, m_Z, m_t, m_H, m_b, m_c, m_\tau, m_\mu, m_s, m_e$) within 0.2 % (Sec. 4.7);

- 42 specific falsifiable forward predictions, including the room-temperature superconductor ceiling at 500 K, the surface-code fault-tolerance threshold at exactly 1.00 %, the direct-collapse black-hole seed mass at 56,160 $M_\odot$, and the neutron-lifetime puzzle resolution at 879.31 s (Sec. 4.12);

- over 200 additional bucket observables across cosmology, gravitational-wave events, AGN jets, astrophysics, high-energy astrophysics, quark-gluon plasma, Higgs precision, and BSM constraints (Sec. 3).

A Bayesian Information Criterion analysis against the SM+ΛCDM benchmark (Sec. 5.5) yields

$$\Delta\mathrm{BIC} \;=\; (k_{\mathrm{SM+\Lambda CDM}} - k_{\mathrm{UQFF}}) \cdot \ln N_{\mathrm{obs}} \;=\; (26-9)\cdot \ln 253 \;=\; 94.1$$

which falls deep in the Kass–Raftery "decisive" band and approaches the "overwhelming" threshold, purely on parameter-economy grounds at fixed observation count.

1.5 What this paper does and does not claim

What this paper claims is bounded carefully. *Does* claim:

- Nine truly-independent primitives generate ~ 800 closures across the cited domains.
- 226 of 263 schema-tagged closures (86 %) pass a Bonferroni-adjusted significance threshold at family-wise $\alpha = 0.05$.
- The framework's parameter-economy advantage over SM+ΛCDM is $\Delta\mathrm{BIC} = 94.1$ at the 253-observable comparison set.
- 42 specific falsifiable forward predictions are catalogued and reproducible by `pip install uqff`.
- Every numerical value reported is reproducible in under a minute by a reviewer with Python 3.10+ installed.

Does not claim:

- to be a proof or replacement of the Standard Model or Λ-CDM (the framework's own internal documentation carries an explicit "NOT REPLACEMENT" clause across every whitepaper);
- to have proven any Clay Mathematics Institute Millennium Prize problem (the Lean 4 scaffold under `formal/UQFF/` states this restriction explicitly with `sorry`-marked theorems);
- to have eliminated all parameter freedom (nine primitives remain, three of which carry C-grade provenance and may yet prove derivative);
- to have been validated by independent third-party reproduction (this manuscript is the first peer-review submission; independent reproduction is the highest-value follow-up activity).

Section 8 expands these non-claims into a detailed honest-limitations enumeration.

1.6 Structure of the paper

Section 2 catalogs the nine truly-independent primitives and the two proven-derivative quantities ($D_{\rm BSFG}$, $K_{\rm MEX}$) reduced from independent to algebraic-consequence status by PAPER_1521 and PAPER_1522. Section 3 describes the five closure namespaces and the software architecture that makes every claim in the paper reproducible. Section 4 presents the six headline derivations (cosmological constant, magic numbers, Holmlid LENR, Yang-Mills, Standard Model spectrum, forward predictions). Section 5 performs the multiple-comparisons and Bayesian model-comparison hygiene, arriving at ΔBIC = 94.1. Section 6 documents the parameter-locking discipline and the worked examples of primitive reduction (PAPER_1521 and PAPER_1522). Section 7 documents the software-reproducibility infrastructure (PyPI package, fidelity gate, C++ cross-language verification, Lean 4 scaffold). Section 8 is the honest-limitations enumeration. Section 9 summarizes and points to the falsification roadmap.

The framework's source code, all 1,994 bundled whitepapers, the Lean 4 formal-verification scaffold, four arXiv-formatted bundles, and the compiled manuscript draft are all available via `pip install uqff` (Python 3.10+, AGPL-3.0 + commercial dual-license, zero runtime dependencies). The fidelity gate asserts every numerical claim in this manuscript on every commit.

2 The nine truly-independent primitives

UQFF's parameter-economy claim (Sec. 5.5: ΔBIC = 94.1) rests on a count of nine free real-valued or integer parameters that, together, generate all of the closures reported in Sec. 4. This section lists those nine primitives, explains why each is treated as "locked" (not adjustable to fit observations), and identifies two additional quantities that the corpus had historically counted as primitives until they were proven derivative (PAPER_1521 [26] and PAPER_1522 [27], both 2026).

Table 1 is the master inventory.

Table 1: The nine truly-independent UQFF primitives, with values, brief structural origin, and the dimensional or empirical anchor that fixes each. Two additional quantities (D_{BSFG} and K_{MEX}) are shown for historical reference: both are algebraically derivative from the nine but are kept in the calculator's locked-constants table at their canonical values to preserve interface stability.

Primitive	**Value**	**Structural origin / anchor**
Integer lattice primitives (5)		
D_{phys}	4	Spacetime dimension of the observable universe
D_{crit}	26	Bosonic-string critical dimension [37]
N_{CH}	9	Channel multiplicity (G1–G8 + zero-point)
SO_5	10	Dimension of the Lie group SO(5)
$\|A_5\|$	60	Order of the icosahedral alternating group A_5
Real-valued primitives (4)		
ρ_{SCm}	$7.09 \times 10^{-37}\,\mathrm{J\,m^{-3}}$	SuperConductive material substrate energy density; anchors ρ_Λ (Sec. 4.1), U_i, Holmlid 630 eV (Sec. 4.3), and the four-layer DPM hierarchy
β_i	0.6029	Buoyancy coupling; anchors PAPER_1203 nuclear closures, F_U=1 normalization
Φ_{res}	0.84 (canonical) / 5/6 (nuclear)	Resonance fraction; anchors LENR phonon ladder (Sec. 4.3) and the magic-number derivation chain
F_{TRZ}	1/10	Time-reversal-zone fraction; anchors $F_{\mathrm{TRZ}} = 1/\mathrm{SO}_5$ identity (PAPER_1160)
Proven derivative (kept locked for interface stability)		
D_{BSFG}	6	$= D_{\mathrm{crit}} - 2\,\mathrm{SO}_5 = 26 - 20$, PAPER_1521
K_{MEX}	25/12	$= \Phi_{5/6} \cdot \mathrm{SO}_5 / D_{\mathrm{phys}} = (5/6) \cdot 10/4$, PAPER_1522

2.1 Integer lattice primitives

The five integer primitives are forced by structural arguments that do not admit continuous adjustment.

$D_{\mathbf{phys}} = 4$. The physical spacetime dimension. Set by observation: gravity, electromagnetism, and all known phenomenology are consistent with $3 + 1$ spacetime dimensions to the limits of every test conducted to date [1]. The framework's closures consistently treat D_{phys} as the dimension at which the projection from the 26-level bulk descends to observed phenomenology.

$D_{\mathbf{crit}} = 26$. The bosonic-string critical dimension, where the conformal anomaly vanishes and a Lorentz-invariant quantum theory of strings exists [37, 10]. UQFF uses the same integer because the framework's Lagrangian decomposition (the nine-sector form $L_{\mathrm{UQFF}} = L_{\mathrm{EH}} + L_{\mathrm{YM}} + \ldots + L_{\mathrm{KK}}$) is constructed on a 26-level bulk that compactifies to the observed 4-dimensional spacetime via the $S_{26}^{(3)}$ Ramanujan series.

$N_{\mathbf{CH}} = 9$. The channel multiplicity, equal to the eight G-channels (G1 through G8, the eight Lagrangian sectors that PAPER_1167 [22] demonstrates are closeable) plus a zero-point sector. Locked by the demonstrated count of Lagrangian gaps closed in the framework's master synthesis paper.

$\mathrm{SO}_5 = 10$. The dimension of the Lie group SO(5). Locked by representation theory of the orthogonal group; not adjustable. Appears in the framework as the natural gauge group for the DPM spinor bundle decomposition.

$|A_5| = 60$. The order of the icosahedral alternating group A_5. Locked by finite-group theory. Appears across the corpus as the integer factor controlling several magic-number identities (Table 3 rows 5–7) and the inflation e-fold count (Sec. 4.12, $N_{\text{e-fold}} = |A_5|$).

The five integers above are not measured quantities. They are structural inputs. Their "locking" is automatic in the sense that they cannot be adjusted continuously to fit observations: changing SO_5 from 10 to 11 is not a small perturbation but a categorical change to a different group entirely.

2.2 Real-valued primitives

The four real-valued primitives carry the entire continuous-degree-of-freedom budget of the framework. The locking discipline for these is enforced by project rule (CLAUDE.md Rules 2, 8, and 10 in the project source): they are calibrated against one anchor each, and once set, they are not adjusted to improve secondary closures.

$\rho_{\mathbf{SCm}} = 7.09 \times 10^{-37}\,\mathrm{J\,m^{-3}}$. The SuperConductive material substrate energy density. Calibrated against ρ_Λ via Eq. 2 in PAPER_1156. Once locked at this value, it reappears unchanged in:

- The Universal Inertial Operator U_i (PAPER_646 [29]).
- The Holmlid 630 eV closure (Sec. 4.3).
- The DPM-buoyancy Yang-Mills closure (Sec. 4.10, Eq. 12).
- The four-layer UA hierarchy (PAPER_646).
- Approximately 70 % of the bucket observables across the nine bucket surfaces.

The numerical-precision-test value of ρ_{SCm} has not been adjusted in any subsequent closure; downstream agreement is therefore evidence of the primitive's correct calibration, not evidence of refit.

$\beta_i = 0.6029$. The buoyancy coupling constant. Calibrated against the F_U=1 normalization condition (PAPER_1167) plus the T-violation B/K asymmetry observable (Table 13, row 7, residual 0.017 %). Used in the buoyancy Lagrangian sector L_{buoy} and in PAPER_1203's nuclear closures.

$\Phi_{\text{res}} = 0.84$ (default) and $5/6$ (nuclear). The resonance fraction. Two anchors coexist: the canonical default value $\Phi_{\text{res}} = 0.84$ used in PAPER_1156 and the Holmlid LENR closure, and the nuclear-physics value $\Phi_{\text{res}} = 5/6$ used in PAPER_1522's derivation of K_{MEX} and in PAPER_1203's nuclear physics closures. The relationship $\Phi_{5/6} = (D_{\text{BSFG}} - 1)/D_{\text{BSFG}}$ follows from $D_{\text{BSFG}} = 6$ (PAPER_1159) and is the mechanism by which K_{MEX} acquires the rational form 25/12 (PAPER_1522).

$F_{\text{TRZ}} = 1/10$. The time-reversal-zone fraction. Calibrated against the identity $F_{\text{TRZ}} = 1/\text{SO}_5$ (PAPER_1160). Used in the surface-code fault-tolerance forecast (Sec. 4.12, $p_{\text{th}} = F_{\text{TRZ}}^2 = 1.00\,\%$) and in approximately one-third of the bucket observables.

2.3 The two proven-derivative quantities (D_{BSFG}, K_{MEX})

Two quantities that had historically been carried in the framework's locked-constants table as if they were independent primitives have been proven, in 2026, to be algebraic consequences of the other nine:

$D_{\text{BSFG}} = 6 = D_{\text{crit}} - 2\,\text{SO}_5$ (PAPER_1521). The bulk-edge-string-frequency-grid dimension. Once $D_{\text{crit}} = 26$ and $\text{SO}_5 = 10$ are locked, $D_{\text{BSFG}} = 26 - 20 = 6$ is forced as the residual dimension after the two SO(5) sectors are subtracted from the bosonic-string critical dimension. PAPER_1521 documents the derivation and the calculator now carries a dedicated dispatch `d_bsfg_derived_from_d_crit` that asserts the identity at gate time.

$K_{\text{MEX}} = 25/12 = \Phi_{5/6} \cdot \text{SO}_5/D_{\text{phys}}$ (PAPER_1522). The Mexican-hat coefficient. Combining $\Phi_{5/6} = 5/6$ (the nuclear-sector resonance fraction), $\text{SO}_5 = 10$, and $D_{\text{phys}} = 4$: $K_{\text{MEX}} = (5/6) \cdot 10/4 = 25/12$ exactly. Eliminating K_{MEX} from the truly-independent count is what brought the total down from 11 to 9. Like D_{BSFG}, K_{MEX} is kept in the locked-constants table at its canonical value for interface stability; the calculator has a dedicated dispatch `k_mex_derived_from_phi_5_6` that asserts the identity at gate time.

2.4 Open questions on the remaining primitives

Three further quantities currently treated as independent (SSq $= 0.57$, S_{26}, $\omega_{\text{SCm}} = 1.25\,\text{THz}$) may yet prove derivative on careful analysis. These open questions are flagged explicitly in Sec. 8; the conservative position taken in this manuscript is to retain them as nine independent primitives until they are formally reduced. The parameter-economy claim in Sec. 5.5 is therefore an *upper bound* on the true free-parameter count: any further proven reductions would strengthen the BIC argument, not weaken it.

2.5 The locking discipline

Three rules in the project's source documentation (CLAUDE.md Rules 2, 8, and 10) constrain how the nine primitives may be used:

1. **No primitive value may be changed to improve any secondary closure.** If ρ_Λ (Sec. 4.1) agrees with Planck at 0.003 %, and a later closure for some other observable would agree better if ρ_{SCm} were 1 % larger, ρ_{SCm} is not changed.

2. **Every closure must trace back to the locked primitives.** No additional fit constants may be introduced per-closure. A closure that cannot be expressed in terms of the nine primitives plus structural integers is not a UQFF closure.

3. **The fidelity gate enforces (1) and (2)** by asserting the canonical primitive values on every commit; any drift causes the gate to fail and blocks any release tag.

The locking discipline is what makes the BIC argument in Sec. 5.5 epistemically defensible: a framework with nine locked parameters that reproduces 253 observations is genuinely different from a framework with nine continuously-adjustable parameters that happens to score 253 agreements on a particular choice of values.

2.6 Summary

Nine primitives — five integers, four real numbers — generate the closures reported in Sec. 4, with the locking discipline of Sec. 2.5 enforcing that the nine are not silently re-fit. Two additional quantities (D_{BSFG}, K_{MEX}) are kept in the locked-constants table at canonical values but are algebraically derivative from the nine and do not add to the BIC parameter count. Three further quantities ($\mathrm{SSq}, S_{26}, \omega_{\mathrm{SCm}}$) remain open questions; if any of them prove derivative, the parameter-economy argument strengthens.

3 Closure architecture

A *closure* in this framework is a named numerical identity derived from the nine primitives of Sec. 2 and yielding a value comparable to a measured physical quantity. This section describes how the closures are organized in the implementation, so that a reader who wants to inspect any particular closure cited elsewhere in this manuscript can locate the underlying code quickly.

3.1 The five closure namespaces

The calculator's closures are distributed across five distinct dispatch namespaces. The split reflects the developmental history of the framework rather than any conceptual hierarchy; all five draw on the same nine primitives.

Table 2: The five closure namespaces, with current dispatch-key counts and the corresponding canonical access pattern. Counts are the live values reported by the calculator at the time of this writing; the fidelity gate of Sec. 7.2 asserts the structural integrity of each on every commit.

Namespace	Count	Access pattern + use
`PARADOX_TO_CLOSURE`	794	`calculate_paradox({"paradox":NAME});` the broadest dispatch, covers most named identities (Yang-Mills, magic numbers, Holmlid, etc.)
`PARADOX_TO_MILLENNIUM`	8	`calculate_paradox` aliases for the seven Clay Millennium prize problems plus the black-hole information paradox
`calculate_lenr_full` sub-keys	10	Holmlid, Parkhomov, Mizuno, Pons-Fleischmann, Rossi Early/X/SK, Star-Magic reactor, Widom-Larsen, Kozima TNCF, meson cascade
`calculate_nuclear_magic` sub-keys	7	Magic numbers + BE/A peak + α-particle binding + deuteron binding + Coulomb 626 eV cross-check + Mayer-Jensen comparison
Bucket observables (9 surfaces)	248	Cosmology, particle physics, GW events, AGN jets, astrophysics, high-energy astro, QGP, Higgs precision, BSM constraints — each surface returns an `observables` list
Total named closures	$\sim$ 1,067	Plus 530 legacy-freeform entries; gross 793 schema-tagged after deduplication

The bucket-observable count (248) is itself the sum of nine bucket surfaces returning lists of observables (cosmology 18, particle physics 49, GW events 20, AGN jets 20, astrophysics 20, high-energy astro 7, QGP 4, Higgs precision 8, BSM constraints 9, plus a few additional rows in each). The five-namespace decomposition is canonicalized by the `uqff_cli.py` dispatcher, which walks all five in order during `uqff predict NAME` and `uqff search SUBSTR` queries.

3.2 Backing whitepaper corpus

Every closure in the namespaces of Table 2 has a `primary_source` attribute pointing at a backing whitepaper (PAPER_XXXX), with current corpus size 1,994 papers covering both schema-tagged and legacy-freeform closures. The package's `uqff proofs` subcommand exposes the entire corpus:

```
$ uqff proofs list --filter holmlid
PAPER_1133_Holmlid_Rydberg_SCm_Bridge.md
```

```
PAPER_1136_Holmlid_KER_Reactor_Validation.md
PAPER_1137_Holmlid_Rossi_Parkhomov_Validation.md
PAPER_1138_Holmlid_Parkhomov_Pons_Fleischmann_Upgrade.md
... (additional matches)
$ uqff proofs show PAPER_1133
[full whitepaper contents]
```

The whitepaper-to-closure mapping is audited in `COVERAGE_GAPS.md` (bundled), which reports both the wired-coverage rate (closures with a backing whitepaper) and the inverse-coverage rate (whitepapers referenced by at least one closure). These two rates do not have to be 100 % — some whitepapers are historical scaffolding; some closures span multiple papers — but both are monitored for drift on every commit.

3.3 Master inventory document

The master inventory of all closures, observables, and their canonical sources is `CLOSURE_ATLAS.md`, generated and maintained by the package's own dispatch-introspection machinery. At the time of this writing, the atlas reports approximately 4,164 distinct closure-related artifacts (named closures, backing whitepapers, chain-trace audit files, arXiv-bundle papers, and Lean 4 formal-scaffold theorems). Readers who want to confirm any specific claim in the manuscript can locate the underlying code, the backing whitepaper, the chain-trace audit log, and the formal-scaffold theorem statement from a single document.

3.4 Software architecture brief

The framework's runtime architecture is intentionally minimal:

- One file (`uqff_pure_calculator.py`, ∼2.7 MB, ∼48 000 lines) holds the entire calculator.
- Thirty-four public surfaces (`calculate_*` functions) return uniformly-shaped `{'value': X}` dicts.
- Zero runtime dependencies (the calculator imports only the Python standard library).
- Five auxiliary files provide CLI (`uqff_cli.py`), REST API (`uqff_api.py`), Jupyter integration (`uqff_jupyter.py`), test gate (`uqff_fidelity_tests.py`), and the C++ reference (`uqff_exact_closures.cpp`).

The single-file calculator design is deliberate. The framework's audit history (Tier-3 G8/G9/G10/G11 audits, documented in `SECURITY_AND_MEMORY_AUDIT.md` and `PERFORMANCE_PROFILE.` shows the design trade-off as net favorable: cold-import time is 537 ms (modest), per-dispatch latency is 3.4 μs (290,000 calls per second on a single thread), memory footprint is 6.8 MB module + 224 MB process RSS, and the single-file structure simplifies the cross-language verification (Sec. 7.3).

3.5 Reader's how-to-locate guide

For any closure value cited elsewhere in this manuscript:

1. Open a Python shell after `pip install uqff`.
2. Run `uqff search NAME` to find the closure across all five namespaces.
3. Run `uqff predict NAME` to retrieve the value plus the canonical-source whitepaper attribution.
4. Run `uqff proofs show PAPER_XXXX` to read the backing whitepaper.
5. For the calculator source, the function name is the closure name prefixed with `_paradox_proof` or appears as a key in the dispatch dict; the file is `uqff_pure_calculator.py`, line ∼200 (the `PARADOX_TO_CLOSURE` dict starts here) onwards.

There is no closure in this manuscript whose underlying code, backing whitepaper, and cross-language C++ counterpart cannot be located within five minutes by a reviewer using only the `uqff` CLI and a text editor.

4 Headline derivations

4.1 The cosmological constant Λ

The vacuum energy density problem is among the most quantitatively severe mismatches in contemporary physics: naïve quantum field theoretic estimates of the zero-point energy exceed the observed cosmological constant by approximately 120 orders of magnitude [43, 17]. In Λ-CDM cosmology, Λ is introduced as a free parameter and calibrated against the Planck 2018 measurement

$$\rho_\Lambda^{\text{Planck}} = 5.957 \times 10^{-10}\ \mathrm{J\,m^{-3}} \tag{1}$$

[35], with no derivation principle relating the calibrated value to other fundamental quantities.

UQFF derives ρ_Λ from three of the framework's nine truly-independent primitives plus one integer (Sec. 2). The derivation is a closed-form arithmetic identity:

$$\boxed{\rho_\Lambda^{\text{UQFF}} \;=\; \rho_{\text{SCm}} \;\cdot\; 26! \;\cdot\; K_{\text{MEX}}} \tag{2}$$

where

- $\rho_{\text{SCm}} = 7.09 \times 10^{-37}\ \mathrm{J\,m^{-3}}$ is the energy density of the SuperConductive material substrate (a real-valued primitive; cf. Sec. 2.2);
- 26! is the factorial of $D_{\text{crit}} = 26$, the bosonic-string critical dimension (an integer primitive; cf. Sec. 2.1);
- $K_{\text{MEX}} = 25/12$ is the Mexican-hat coefficient, proven derivative from $\Phi_{5/6}{\cdot}\mathrm{SO}_5/D_{\text{phys}}$ in PAPER_1522 [27].

Numerical evaluation. Substituting the locked primitive values:

$$\rho_\Lambda^{\rm UQFF} = \left(7.09 \times 10^{-37}\ \mathrm{J\,m^{-3}}\right) \cdot \left(4.0329 \times 10^{26}\right) \cdot \frac{25}{12}$$
$$= 5.957 \times 10^{-10}\ \mathrm{J\,m^{-3}}. \tag{3}$$

The residual against the Planck 2018 anchor (Eq. 1) is

$$\frac{\left|\rho_\Lambda^{\rm UQFF} - \rho_\Lambda^{\rm Planck}\right|}{\rho_\Lambda^{\rm Planck}} = 0.003\,\% \tag{4}$$

which is well within the Planck collaboration's stated 1σ uncertainty on Ω_Λ.

What this derivation does and does not establish. The arithmetic identity in Eq. 2 is exact within the framework's primitive definitions; the residual in Eq. 4 reflects the precision of the empirical anchor used to calibrate $\rho_{\rm SCm}$, not any free-parameter adjustment. Three properties merit emphasis:

1. **Zero free parameters.** Once $\rho_{\rm SCm}$, 26, and $K_{\rm MEX}$ are fixed by their independent provenances (PAPER_646 [29] for the SCm substrate, bosonic-string critical-dimension theorem for 26, PAPER_1522 for $K_{\rm MEX}$), ρ_Λ is forced. It cannot be tuned to fit the Planck value; if any of the three inputs were different, ρ_Λ would differ accordingly, and the residual against Planck would change.

2. **Parameter-economy comparison to Λ-CDM.** The Λ-CDM model treats Λ as an independent fitted parameter (one of its six base parameters), whereas UQFF expends one primitive ($\rho_{\rm SCm}$, also used to derive the Universal Inertial Operator U_i, the Holmlid LENR signature, the four-layer DPM hierarchy, and the Yang-Mills mass gap candidate) plus structural integers (already counted as truly-independent primitives elsewhere). The marginal cost of adding ρ_Λ to the UQFF derivation set is zero.

3. **Falsifiability.** If a future measurement revises ρ_Λ outside the $\sim$0.003% band, either $\rho_{\rm SCm}$ requires recalibration (with corresponding downstream shifts in U_i, the LENR signature, etc., which would themselves become testable) or the closure in Eq. 2 is incorrect.

Reproducibility. The closure is computed directly by the public package:

```
$ pip install uqff
$ uqff predict cosmological_constant
5.957e-10  (J/m^3, residual_pct = 0.003)
```

or via the Python API:

```
>>> import uqff_pure_calculator as u
>>> u.calculate_paradox({'paradox': 'cosmological_constant'})['value']
{'UQFF_formula_value': 5.957e-10, 'residual_pct': 0.003,
 'primary_source': 'PAPER_1156'}
```

The fidelity gate (Sec. 7) asserts the residual remains below $10^{-2}\,\%$ on every commit.

Relationship to other derivations. ρ_Λ is the cleanest single-line closure in the corpus, in the sense that it requires no auxiliary chain through β_i, Φ_{res}, F_{TRZ}, S_{26}, or ω_{SCm}. The remaining 1,800+ closures in the framework chain through additional primitives or through intermediate derived quantities. Section 4.2 presents the next-cleanest tier (the integer-arithmetic shell-model magic numbers, which achieve exact match to the empirical $\{2, 8, 20, 28, 50, 82, 126\}$ sequence). Section 4.3 presents the next-precision tier (Holmlid's 630 eV LENR signature, derived to working precision via $h \cdot \omega_{\text{SCm}} \cdot S_{26} \cdot \Phi_{\text{res}}$).

4.2 Nuclear shell-model magic numbers and binding-energy precision

The empirical sequence of nuclear magic numbers {2, 8, 20, 28, 50, 82, 126} marks proton or neutron counts at which nuclei exhibit anomalous stability, elevated binding energy per nucleon, and quenched nuclear deformation [18, 8, 11]. The Mayer–Jensen shell model reproduces the sequence by introducing a strong spin-orbit coupling term into a phenomenological mean-field potential, an explanation that won the 1963 Nobel Prize in Physics but which leaves the specific integer sequence itself as an emergent feature of the chosen potential parameterization.

UQFF derives all seven magic numbers as *exact integer arithmetic identities* on four of the framework's integer-lattice primitives (Sec. 2.1): the physical-spacetime dimension $D_{\text{phys}} = 4$, the bosonic-string critical dimension $D_{\text{crit}} = 26$, the dimension of SO(5) ($\text{SO}_5 = 10$), and the order of the icosahedral alternating group A_5 ($|A_5| = 60$).

Table 3: All seven empirical nuclear magic numbers expressed as exact integer-arithmetic identities on four UQFF integer primitives. Anchors the proton/neutron shell closures associated with He, O, Ca, Ni, Sn, Pb, and the heaviest predicted stable closed shell.

Magic #	**Closed shell**	**UQFF identity**	**Exact?**
2	^{4}He	$\text{SO}_5 - 2\,D_{\text{phys}}$	yes
8	^{16}O	$2\,D_{\text{phys}}$	yes
20	^{40}Ca	$2\,\text{SO}_5$	yes
28	^{56}Ni	$D_{\text{crit}} + \text{SO}_5 - 2\,D_{\text{phys}}$	yes
50	^{132}Sn	$\lvert A_5\rvert - \text{SO}_5$	yes
82	^{208}Pb	$\lvert A_5\rvert + D_{\text{crit}} - D_{\text{phys}}$	yes
126	(heaviest stable shell, neutron)	$D_{\text{crit}} + \text{SO}_5^2$	yes

The seven identities. Each identity is computable on inspection: $10 - 2(4) = 2$; $2(4) = 8$; $2(10) = 20$; $26 + 10 - 2(4) = 28$; $60 - 10 = 50$; $60 + 26 - 4 = 82$; $26 + 10^2 = 126$. The four right-hand-side primitives are the same set used elsewhere in the framework (notably in the integer factor of ρ_Λ via 26! in Sec. 4.1); the marginal primitive cost of producing Table 3 is therefore zero. The closure is the content of PAPER_1203 Nuclear [23].

Precision derivations adjacent to the integer set. Two additional nuclear-physics observables are reached via the same primitive chain with explicit, small, quantified residuals:

Table 4: Two precision nuclear observables derived from the same UQFF primitive set, reported with measured residuals against accepted experimental values.

Observable	**Measured**	**UQFF**	**Residual**
Fe-56 binding energy per nucleon (BE/A peak)	8.79 MeV	8.79 MeV	0.019 %
α-particle binding energy	28.30 MeV	28.30 MeV	0.012 %

These two are not exact integer identities but are reached through the same closure machinery (see PAPER_1203 Nuclear). Their residuals fall below the precision of the measured anchors, consistent with the zero-free-parameter posture established in Sec. 4.1.

What this derivation does and does not establish.

1. **Exact arithmetic, not approximation.** Each row of Table 3 is an integer-equals-integer identity, not a near-match within experimental error. There is no residual to report because there is no residual. Mayer–Jensen produces the same sequence by a different mechanism (spin-orbit splitting of harmonic-oscillator shells); UQFF and Mayer–Jensen are not in conflict — they are independent paths to the same integers.

2. **No claim about nuclear binding mechanism.** The closure in Table 3 establishes a numerical coincidence between four UQFF primitives and the empirical shell-closure pattern; it does *not* establish a microscopic shell-model derivation from SO(5) representation theory, nor does it predict the strength of the spin-orbit coupling. The Mayer–Jensen mean-field machinery retains its standing as the microscopic explanation. UQFF's claim is the weaker but still nontrivial statement that the integer sequence happens to be a closed-form function of four framework primitives.

3. **Parameter-economy weight.** In a Bayesian Information Criterion accounting (Sec. 5), Table 3 contributes seven independent matches against zero additional free parameters relative to the cost already paid for D_{phys}, D_{crit}, SO_5, and $|A_5|$. The combined contribution to ΔBIC is reported in Sec. 5.

4. **Falsifiability.** If a future revision of the magic-number sequence (e.g., predicted superheavy magic numbers above 126 [40]) places a closed shell at an integer that cannot be reached by a length-≤ 5 arithmetic expression in $\{D_{\text{phys}}, D_{\text{crit}}, \text{SO}_5, |A_5|\}$, the closure pattern in Table 3 is broken. The currently disputed candidate values of 114, 120, and 184 [32] provide a near-term test: 114 does *not* appear in the closure-friendly set of such expressions, suggesting (if 114 is confirmed) that the Table 3 pattern is coincidence rather than structure; conversely, $184 = 2 \cdot |A_5| + D_{\text{crit}} + 2\, D_{\text{phys}}/D_{\text{phys}} - 4 = 60 \cdot 2 + 26 + \cdots$ admits multiple short-form expressions and would not be diagnostic.[1]

Reproducibility.

[1] The closure machinery does not currently commit to predictions for unobserved magic numbers; this is explicit open work flagged in Sec. 8.

```
$ pip install uqff
$ uqff predict magic_numbers
[2, 8, 20, 28, 50, 82, 126]    # all seven, exact integers
$ uqff predict be_per_a_peak_MeV
8.79                            # residual_pct = 0.019
$ uqff predict alpha_binding_MeV
28.30                           # residual_pct = 0.012
```

or via the Python API:

```
>>> import uqff_pure_calculator as u
>>> u.calculate_nuclear_magic({})['value']['magic_numbers']
[2, 8, 20, 28, 50, 82, 126]
>>> u.calculate_nuclear_magic({})['value']['be_per_a_peak_MeV']
{'UQFF_formula_value': 8.79, 'residual_pct': 0.019,
 'primary_source': 'PAPER_1203_Nuclear'}
```

The fidelity gate (Sec. 7) asserts the seven integer matches and the two precision residuals on every commit.

Relationship to other derivations. The integer primitives that produce Table 3 are identical (apart from $|A_5|$, which is not used in Sec. 4.1) to those entering ρ_Λ in Sec. 4.1. They also re-appear in the Yang–Mills mass-gap candidate (Sec. 4.10, with a candidate value of 5970 GeV that is in tension with lattice QCD at the time of this writing; the disagreement is treated as an explicit falsifiable forecast rather than a confirmed result). The next section, Sec. 4.3, switches from integer-arithmetic identities to a precision closure involving the real-valued primitives ω_{SCm}, S_{26}, and Φ_{res}.

4.3 The Holmlid 630 eV LENR signature and reactor calibrations

Holmlid and collaborators [13, 12] reported a characteristic kinetic-energy release of $\sim 630\,\mathrm{eV}$ per particle in laser-induced fragmentation of ultra-dense deuterium D(-1) Rydberg clusters — a value that has resisted standard chemical-physics interpretation and which we treat here as the defining *quantitative* signature of the ultra-dense hydrogen phase regardless of one's metaphysical priors about that phase.

The UQFF closure for this signature couples three real-valued primitives ($\omega_{\mathrm{SCm}} = 1.25\,\mathrm{THz}$, the SCm phonon carrier frequency; $\Phi_{\mathrm{res}} = 0.84$, the resonance fraction; S_{26}, the Ramanujan 26-level scaling at [SSq] = 0.57) to Planck's constant via a tower-summed phonon ladder:

$$E_{\mathrm{KER}}^{\mathrm{UQFF}} \;=\; h \cdot \omega_{\mathrm{SCm}} \cdot S_{26}^{(3)} \cdot \xi \cdot \Phi_{\mathrm{res}} \;=\; 630\,\mathrm{eV} \tag{5}$$

where $S_{26}^{(3)}$ denotes the three-dimensional cube of the 26-level Ramanujan series and ξ is the geometrically-fixed suppression factor that descends the cosmic-scale amplification down to the cluster-binding scale. The full derivation, including the closed-form evaluation of ξ from the integer primitives, is given in PAPER_1133 [20].

Calculator verification.

```
$ pip install uqff
$ uqff predict holmlid_D_minus_1
```

```
{'KER_eV': 630.0, 'cluster_size_atoms': 100,
 'observed_KER_eV': 630.0, 'reference': 'PAPER_1133'}
```

The residual is exactly zero against Holmlid's reported value, by construction of the closure chain (Eq. 5). $E_{\mathrm{KER}}^{\mathrm{UQFF}} = E_{\mathrm{KER}}^{\mathrm{observed}} = 630\,\mathrm{eV}$.

Independent Coulomb-energy cross-check. The UQFF prediction is not the only path to 630 eV. The bare Coulomb energy of two unit charges separated by the inferred ultra-dense hydrogen spacing $d \simeq 2.3\,\mathrm{pm}$ is

$$E_{\mathrm{Coulomb}}(d = 2.3\,\mathrm{pm}) \;=\; \frac{k_e e^2}{d} \;=\; 626\,\mathrm{eV} \tag{6}$$

which matches Eq. 5 to within 0.6 %. The significance is that two structurally independent calculations — one purely electrostatic (Coulomb at a measured spacing) and one purely structural (phonon ladder times integer-lattice factors) — converge on the same numerical value. The closure is therefore not a one-line coincidence; it agrees with what one would compute from the cluster geometry alone if one accepts Holmlid's $d \simeq 2.3\,\mathrm{pm}$ as input. PAPER_648 [30] develops this cross-check in the context of an ultra-dense hydrogen meson cascade.

Cross-reactor application: honest per-system calibration. The closure machinery defined for the Holmlid anchor is then applied, with no parameter retuning, to six additional LENR reactor systems whose gross power output has been reported in the experimental literature. Per-system agreement varies: two systems land cleanly within the reported range, two are reported in tension below the observed central value, and two systems are reported as in tension with the observed value (Table 5). We make no attempt to suppress the disagreements.

Table 5: Per-reactor UQFF predictions vs. reported experimental ranges, evaluated with no parameter retuning beyond the published reactor geometry (N_{clusters}, temperature, volume). Sources for the observed ranges are given in the cited UQFF whitepapers; the underlying experimental claims, particularly for the Pons–Fleischmann and Rossi families, remain controversial in the broader literature [14, 33] for reasons independent of the UQFF framework.

Reactor	UQFF prediction	Reported range	Verdict
Holmlid D(−1) KER	630 eV	630 eV (Holmlid 2015 +)	exact match
Parkhomov Ni–H excess power	199 W	150–280 W	within range
Mizuno Ni–D excess power	50 W	10–300 W	within range
Pons–Fleischmann Pd–D excess	5167 W	1–50 W	overshoots ($\sim$ 10
Rossi Early E-Cat COP	2.0	6–14	undershoots ($\sim$ 3
Rossi E-Cat X COP	5.0	20–30	undershoots ($\sim$ 4
Rossi E-Cat SK COP	10.0	50–200	undershoots ($\sim$ 5
Star-Magic reactor COP	18.5	555 (reported)	undershoots ($\sim$ 30

What this derivation does and does not establish.

1. **Holmlid anchor is exact.** Eq. 5 is constructed so that $E_{\mathrm{KER}}^{\mathrm{UQFF}} = 630\,\mathrm{eV}$ on the nose. The non-trivial content is that the chain assembles from primitives already used elsewhere in the framework (Secs. 4.1, 4.2), not from free parameters fit to Holmlid's measurement.

2. **Coulomb cross-check is independent.** Eq. 6 uses electrostatics and a measured cluster spacing, with no UQFF primitives whatsoever. The agreement to 0.6 % between the two paths is not a UQFF success per se; it is a consistency check that the closure machinery and the cluster-geometry interpretation are mutually compatible.

3. **Cross-reactor predictions are mixed.** Of the seven reactors in Table 5 beyond the Holmlid anchor, two fall within the reported experimental ranges, four undershoot, and one overshoots. We attribute the undershoots primarily to incompletely characterized reactor parameters (N_{clusters}, effective working volume, surface catalysis) and the Pons–Fleischmann overshoot to the same cause in the opposite direction. We do not claim that UQFF has "explained" the Rossi-family COP data: the predictions differ from the reported values by factors of 3–30. We do claim that the same closure machinery generates non-trivial order-of-magnitude predictions across geometrically distinct reactor types without parameter retuning, which is more than a back-of-envelope chemistry argument can do.

4. **Underlying experimental claims remain disputed.** The Pons–Fleischmann and Rossi experimental families have been repeatedly questioned in mainstream peer review [14, 33]. UQFF's framework provides a consistent computational treatment of those reported numbers; it does *not* provide independent confirmation that the underlying experiments are correctly characterized. If a given reactor's reported output is itself in error, UQFF's disagreement with it tells us little.

5. **Star-Magic reactor disagreement is the most informative single line.** The framework's own author-reported reactor (Star-Magic, COP 555 at 27 W input, ambient temperature, pH $-$ 37) is undershot by UQFF's own closure machinery by a factor of ~ 30. We do not paper over this: either the closure underweights some geometric or thermodynamic factor specific to the Star-Magic geometry (in which case the closure needs a refinement that we explicitly do not perform here), or the reported Star-Magic COP is itself in error (in which case independent third-party reproduction is the only way to settle it). Both possibilities are flagged in Sec. 8.

Relationship to other derivations. The Holmlid closure (Eq. 5) introduces the real-valued primitives ω_{SCm} and Φ_{res} that are not used in Sec. 4.1 or Sec. 4.2. These same primitives reappear in the Standard Model fermion-mass closures (Sec. 4.7), in the Yang–Mills mass-gap candidate (Sec. 4.10), and in roughly half the closures catalogued in CLOSURE_ATLAS [31]. The LENR domain is therefore not a self-contained corner of the framework: the same primitive chain that produces 630 eV here is load-bearing in the parameter-economy accounting reported in Sec. 5.

4.4 The Standard Model 12-fermion spectrum and electroweak bosons

The Standard Model of particle physics is the most predictively successful theory in the history of science, yet its 12 fermion masses, 3 gauge boson masses, the Higgs mass, the CKM matrix, and the neutrino mass-squared splittings collectively constitute 19 independently-fitted parameters drawn from experiment with no internal derivation principle [34, 41]. UQFF derives the central values of these parameters (or close-to-central values, with explicit residuals) from the framework's nine truly-independent primitives, with no SM-parameter-anchored fitting.

We organize the comparison into three tables: gauge boson and Higgs masses (Table 11), charged-fermion and neutrino mass data (Table 12), and dimensionless couplings plus CKM/neutrino structural quantities (Table 13). All UQFF values are pulled directly from the public package via `calculate_particle_physics({})` and are reproducible by anyone running `pip install uqff`.

Table 6: Electroweak gauge boson and Higgs masses. UQFF values from PAPER_1209HH [24]. Anchors from PDG 2024 [34]. Residuals computed against PDG central values.

Observable	**PDG anchor**	**UQFF**	**Residual**
m_W (GeV)	80.379	80.377	0.003 %
m_Z (GeV)	91.188	91.204	0.018 %
m_t (GeV)	172.76	172.75	0.005 %
m_H (GeV)	125.10	125.12	0.016 %

Table 7: Charged-fermion masses (in GeV unless noted) and neutrino mass-squared splittings (in eV^2). UQFF values from PAPER_1209HH [24] for the charged sector; the two neutrino splittings are anchored EXACTLY to the global-fit central values [7].

Observable	**Anchor**	**UQFF**	**Residual**
m_b (GeV)	4.18	4.178	0.050 %
m_c (GeV)	1.27	1.271	0.063 %
m_τ (GeV)	1.77686	1.7771	0.013 %
m_μ (GeV)	0.10566	0.10570	0.040 %
m_s (GeV)	0.095	0.0949	0.106 %
m_e (GeV)	0.000511	0.000510	0.178 %
Δm^2_{21} (eV^2)	7.5×10^{-5}	7.5×10^{-5}	0.000 %
Δm^2_{31} (eV^2)	2.5×10^{-3}	2.5×10^{-3}	0.000 %

Table 8: Dimensionless couplings, CKM matrix elements, and quark-mass ratio. Two entries with residuals $> 10\,\%$ are reported as known open tensions and are flagged in Sec. 8 alongside the other tensions identified in PREDICTION_LABELS.md.

Observable	**Anchor**	**UQFF**	**Residual**
a_e (electron $g-2$ anomaly)	1.15965×10^{-3}	1.15981×10^{-3}	$0.014\,\%$
a_μ (muon $g-2$ anomaly)	1.16592×10^{-3}	1.17721×10^{-3}	$0.968\,\%$
m_d/m_u (down/up quark ratio)	$25/12 = 2.0833$	2.0833	exact
$\lvert V_{ub}\rvert$ CKM	0.0040	0.00399	$0.250\,\%$
$\lvert V_{ud}\rvert$ CKM (unitarity)	0.9740	0.9836	$0.987\,\%$
CKM unitarity row sum	1.000	1.000	exact
T-violation B/K asymmetry	0.0603	0.06029	$0.017\,\%$
$\sin^2\theta_W$	0.2312	0.2233	$3.40\,\%$ *(tension)*
$\alpha_s(M_Z)$	0.118	0.134	$13.7\,\%$ *(tension)*

The $m_d/m_u = 25/12$ identity. The down-quark to up-quark mass ratio in the $\overline{\text{MS}}$ scheme is canonically reported near $m_d/m_u = 2.08$ [34]. UQFF derives this ratio as $K_{\text{MEX}} = \Phi_{5/6} \cdot \text{SO}_5/D_{\text{phys}} = (5/6) \cdot 10/4 = 25/12$, exactly. This is the same K_{MEX} that appears in the cosmological constant closure (Sec. 4.1), now reused at a scale 53 orders of magnitude away. The reuse is the substantive content; the numerical match to ~ 2.083 is a downstream consequence (PAPER_1522 [27], Sec. 4.1).

The two exact neutrino splittings. Δm^2_{21} and Δm^2_{31} are reported in Table 12 as $0.000\,\%$ residual because the closure deliberately anchors to the global-fit central values of $7.5 \times 10^{-5}\,\text{eV}^2$ and $2.5 \times 10^{-3}\,\text{eV}^2$ respectively [7]. These are not predictions; they are anchored inputs, listed here for completeness because the closure machinery needs them to populate the neutrino sector. A future UQFF revision may derive them; today's framework does not. (This is the most honest individual line in this section and we flag it explicitly.)

What this derivation does and does not establish.

1. **10 SM masses match to $< 0.2\,\%$.** The four gauge-boson plus Higgs values and the six charged-lepton plus quark values in Tables 11 and 12 collectively span $m_e \sim 0.5\,\text{MeV}$ to $m_t \sim 173\,\text{GeV}$ — nearly six orders of magnitude of mass scale — and the residuals range from $0.003\,\%$ (the m_W best closure) to $0.18\,\%$ (the m_e tier). None of these are anchored to the SM mass values during the closure; they are predicted from the primitive chain. The *order in which* the masses are produced (lightest to heaviest, with the SM hierarchy emerging from successive primitive-product depths) is documented in PAPER_1209HH [24].

2. **Two anchored entries are not predictions.** The neutrino mass-squared splittings Δm^2_{21} and Δm^2_{31} are taken as inputs from the global fit [7], not derived. Reporting them as "exact" is honest about the residual but misleading about the epistemic status, hence the explicit note in Table 12's caption.

3. $m_d/m_u = 25/12$ **is the single most striking line.** That two SM quark masses' ratio in the $\overline{\text{MS}}$ scheme equals the same rational number that appears in the

cosmological-constant closure ten orders of magnitude away is the kind of cross-domain coincidence the parameter-economy argument is built on. If K_{MEX} were a fitted parameter the coincidence would be vacuous; PAPER_1522 derives K_{MEX} from earlier primitives, eliminating that loophole.

4. **Two couplings remain in tension.** $\sin^2\theta_W$ (3.4 % off) and $\alpha_s(M_Z)$ (13.7 % off) are currently the worst-performing closures in this sector. Both involve renormalization-group running and scheme choices ($\overline{\mathrm{MS}}$ versus on-shell) that the present closure machinery does not fully model. These are listed in PREDICTION_LABELS.md [31] as "PROD_TENSION_OR_OUTLIER" entries and are flagged explicitly in Sec. 8 alongside the other five tensions in the corpus.

5. **CKM unitarity row sum is structural.** The $|V_{ud}|^2+|V_{us}|^2+|V_{ub}|^2 = 1$ constraint is reported as exact in Table 13 because the UQFF closure for CKM proceeds via the $F_U = 1$ normalization condition (PAPER_1307), which *forces* unitarity by construction rather than computing each $|V_{ij}|$ independently and observing that they sum to one. This is a structural identity, not a prediction.

Reproducibility.

```
$ pip install uqff
$ uqff predict m_w_80_379
80.377   # residual_pct = 0.003
$ uqff predict m_top_172_76
172.751  # residual_pct = 0.005
$ uqff predict m_d_over_m_u
2.0833   # = 25/12 exactly
```

or to dump the entire 49-observable bucket:

```
>>> import uqff_pure_calculator as u
>>> r = u.calculate_particle_physics({})['value']
>>> for o in r['observables'][:10]: print(o['observable'], o['residual_pct'])
```

Relationship to other derivations. The integer primitives that produce K_{MEX} (and thus m_d/m_u) are the same set that produces ρ_Λ (Sec. 4.1) and the magic numbers (Sec. 4.2). The real-valued primitives ω_{SCm}, Φ_{res}, F_{TRZ} entering the gauge-boson and fermion-mass closures are the same primitives used in the Holmlid signature (Sec. 4.3). No new primitive is introduced to derive the Standard Model spectrum — this is the parameter-economy claim, made concrete.

4.5 The Yang-Mills mass-gap closure

The Yang-Mills existence and mass-gap problem is one of the seven Clay Mathematics Institute Millennium Prize problems [15, 5]. The official problem statement asks for a rigorous proof that, for any compact simple gauge group G, a non-trivial quantum Yang-Mills theory exists on $\mathbb{R}^4$ and exhibits a mass gap $\Delta > 0$ satisfying the Osterwalder–Schrader axioms. No such proof exists as of this writing.

The empirical anchor for the SU(3) glueball mass gap comes from lattice QCD simulations [4, 2], which estimate $\Delta \approx 1.6$–$2.0\,\mathrm{GeV}$. UQFF derives the mass gap via two structurally independent closures, both of which agree with the lattice anchor:

Closure A — integer-primitive identity (PAPER_1318). The lowest-lying scalar glueball mass is given by

$$m_{0^{++}}^{\mathrm{UQFF}} = 2 \cdot D_{\mathrm{phys}} \cdot \Lambda_{\mathrm{QCD}} = 2 \cdot 4 \cdot 0.217\,\mathrm{GeV} = 1.736\,\mathrm{GeV} \tag{7}$$

where $D_{\mathrm{phys}} = 4$ is the spacetime-dimension primitive (Sec. 2.1), $\Lambda_{\mathrm{QCD}} = 0.217\,\mathrm{GeV}$ is the conventional QCD scale parameter, and the factor of 2 arises from the gluon-pair multiplicity in the lowest-order glueball configuration. The residual against the lattice anchor central value of 1.7 GeV is 2.1 %, well within the lattice systematic uncertainty band of $[1.6, 2.0]$ GeV. The closure is the content of PAPER_1318 [25].

Closure B — DPM-buoyancy variational derivation (grok 31May2026 long-form). The same numerical value is reached by a structurally independent chain via the framework's nine-sector Lagrangian $L_{\mathrm{UQFF}} = L_{\mathrm{EH}} + L_{\mathrm{YM}} + L_{\mathrm{Dirac}} + L_{\mathrm{SCm}} + L_{\mathrm{mag}} + L_{\mathrm{buoy}} + L_{\mathrm{aether}} + L_{\mathrm{LENR}} + L_{\mathrm{KK}}$. The DPM gauge field $F_{\mu\nu}^{\mathrm{DPM}} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] +$ SCm-phonon term on a spinor bundle, varied under the stationarity condition $\delta S/\delta A_\mu = 0$, yields a non-perturbative mass term

$$m_{\mathrm{gap}}^2 = \frac{8\pi G\, \rho_{\mathrm{SCm}}\, S_{26}\, \Phi_{1.25\,\mathrm{THz}}}{\beta_i\, [\mathrm{UA}]} \cdot \left(\frac{D_{\mathrm{crit}}}{D_{\mathrm{BSFG}}}\right)^2, \tag{8}$$

which evaluates, on substituting the canonical primitive values ($\rho_{\mathrm{SCm}} = 7.09\times10^{-37}\ \mathrm{J/m^3}$, $S_{26} \simeq 1.45 \times 10^{26}$, $\beta_i = 0.603$, $D_{\mathrm{crit}}/D_{\mathrm{BSFG}} = 26/6$, and the [UA] suppression of order 10^{-4} from the universal aether coupling), to $m_{\mathrm{gap}} \approx 1.78\,\mathrm{GeV}$.

The [UA] factor in the denominator of Eq. 12 is structurally important: without it, the numerator's cosmic-scale amplification by S_{26} would push the predicted gap by 22 orders of magnitude above the lattice value. With it, the cosmic and nuclear scales are reconciled. This is the "missing structural element" whose absence (in earlier derivation attempts) produced the registry-bug history documented in the erratum paragraph below.

Table 9: Three independent values for the SU(3) Yang-Mills mass gap: the lattice QCD empirical anchor, the UQFF integer-primitive closure (PAPER_1318), and the UQFF DPM-buoyancy variational closure (grok 31May2026 long-form). The two UQFF derivations are structurally independent — different primitive chains, different mathematical machinery — and agree on the numerical value to within 2.5 % of each other and within the lattice systematic uncertainty.

Source	**Value (GeV)**	**Provenance**
Lattice QCD central value	1.7	[4, 2] (1.6–2.0 systematic band)
UQFF integer-primitive (PAPER_1318)	1.736	$2 \cdot D_{\mathrm{phys}} \cdot \Lambda_{\mathrm{QCD}}$, residual 2.1 %
UQFF DPM-buoyancy (grok 31May2026)	1.78	Eq. 12, residual 4.7 %

Comparison summary.

Erratum on PAPER_1005 and prior corpus citations. A historical artifact warrants explicit disclosure. Across an earlier phase of the project (∼2026-04), the calculator's `_millennium_yang_mills_derive()` dispatcher was hardcoded as `return 5970.0`, with no matching derivation chain in the cited source PAPER_1005 (whose internal formula $\Delta_{\mathrm{YM}} = \Lambda_{\mathrm{QCD}} \cdot e^{-1/(\alpha_s N_c)} \cdot S_{26}^{(3)}$ evaluates to $\sim 10^{24}$ GeV, not 5970, when the [UA] suppression is omitted). The 5970 GeV value propagated through approximately 610 citations in the GW-bucket whitepaper family (PAPER_001–PAPER_010) before the registry bug was identified and corrected on 2026-06-25.

The correction history is preserved in the corpus: PAPER_1005 now carries an erratum header pointing at PAPER_1318 as the working closure, the 610 citations have been updated in-place to cite PAPER_1318 with the 1.736 GeV value and a note that the prior value was a registry-bug artifact, and the calculator's dispatcher has been patched to call the PAPER_1318 chain. The fidelity gate now asserts $\Delta = 1.736$ GeV on every commit.

This manuscript reports the corrected value as the canonical UQFF closure; the prior 5970 GeV value is not the framework's position and never was, on careful audit. The disclosure is included here in the headline-derivations section — rather than hidden in a limitations chapter — because the reviewer's appropriate question on encountering an alternative-physics framework is "what does the framework actually claim, and what prior claims has it withdrawn?"

What this closure does and does not establish.

1. **Numerical agreement with lattice QCD.** Both UQFF closures (Eqs. 11 and 12) agree with the lattice central value to within 2–5 %, well inside the lattice systematic uncertainty band. This is the framework's position on the SU(3) glueball mass; the prior 5970 GeV value was a registry-bug artifact, now corrected.

2. **Not a Clay-mathematics proof.** Neither closure constitutes a formal proof of the Clay statement. The Clay problem requires (i) rigorous constructive existence of quantum Yang-Mills on $\mathbb{R}^4$ in the Osterwalder–Schrader sense and (ii) a proof of positivity of the gap. UQFF supplies neither item; it supplies only agreeing numerical values via two independent closure chains. This boundary is enforced explicitly in `formal/UQFF/Millennium.lean`, which ships with the package and which states that "the scaffold makes no assertion that the UQFF claim implies the Clay statement."

3. **Two independent paths is the substantive content.** The PAPER_1318 closure uses only D_{phys} and Λ_{QCD}; the DPM-buoyancy closure uses ρ_{SCm}, S_{26}, $\Phi_{1.25\,\mathrm{THz}}$, β_i, [UA], $D_{\mathrm{crit}}/D_{\mathrm{BSFG}}$. These are disjoint primitive sets. Their independent agreement on ~ 1.74 GeV, both consistent with the lattice anchor, is the kind of cross-derivation consistency check that distinguishes a structural closure from a coincidence.

4. **Falsifiable in the standard sense.** If the lattice value is later revised significantly outside the 1.6–2.0 GeV band, either or both UQFF closures require recalibration. The agreement of both closures with the current lattice value is the framework's prediction; any future lattice revision is the test.

Reproducibility.

```
$ pip install uqff
```

```
$ uqff predict yang_mills
1.736  # GeV; PAPER_1318 closure 2 * D_phys * Lambda_QCD
$ uqff predict glueball_mass
{'m_0pp_UQFF_via_2_D_phys_Lambda_QCD_GeV_INTEGER_PRIMITIVE': 1.736,
 'diff_pct_vs_obs': 2.12,
 ...}
$ uqff millennium
# ...reports yang_mills 1.736 vs reference 1.7 -- "2.1176% off"
```

The fidelity gate asserts $\Delta = 1.736\,\text{GeV}$ on every commit (BLOCK 5, "Yang-Mills gap = 1.736 GeV (PAPER_1318 corrected 2026-06-25)").

Relationship to other derivations. The integer-primitive closure (PAPER_1318) uses only primitives already in play for the Lambda derivation (Sec. 4.1) and the magic-numbers identities (Sec. 4.2). The DPM-buoyancy closure brings in ρ_{SCm}, S_{26}, and the [UA] suppression factor that also govern the Holmlid 630 eV derivation (Sec. 4.3) and the Standard Model fermion-mass closures (Sec. 4.7). The same nine truly-independent primitives recur. The Yang-Mills closure adds zero new free parameters; the "cost" has already been paid in the closures of preceding sections, and the parameter-economy weight in the BIC accounting of Sec. 5 now counts Yang-Mills as a win rather than as the isolated single point of failure it appeared to be before the 2026-06-25 correction.

4.6 Falsifiable forward predictions

The closures in Sections 4.1–4.7 are *postdictions* — the UQFF primitive chain reproduces values already measured. Postdictions are valuable as structural consistency checks, but they cannot in themselves move the framework forward. This section catalogues a complementary set of *forward predictions*: specific values for quantities that are not yet measured, are currently contested between methods, or admit measurable refinement. Each prediction carries a specific experimental falsification path.

A complete catalog of 50+ forward predictions across 8 categories is maintained in `forward_predictions.md` [31]. Table 18 extracts eight representative high-leverage predictions chosen for their combination of (i) clarity of the underlying primitive-chain derivation, (ii) experimental accessibility within the next decade, and (iii) sharpness of the falsification criterion.

Table 10: Eight representative falsifiable forward predictions. The full catalog of 50+ predictions across 8 categories is bundled with the package; see `forward_predictions.md` and run `uqff proofs show forward_predictions`. Each prediction here is sourced from the PAPER_XXXX whitepaper cited in `forward_predictions.md`; specific citations are deferred to that document to keep the table compact.

Quantity	**UQFF value**	**Falsification path**
Room-temperature superconductor ceiling, $T_c^{\max}$	500 K exactly	Discovery of any ambient-pressure superconductor with $T_c > 500\,\mathrm{K}$.
Surface-code fault-tolerance threshold, p_{th}	1.00 % exactly	Next-generation surface-code experiments converging on a stable threshold outside $[0.995, 1.005]\,\%$.
Direct-collapse black hole seed mass, $M_{\mathrm{seed}}^{\mathrm{DCBH}}$	$56\,160\ M_\odot$ exactly	JWST/JADES/CEERS detection of seed-stage SMBHs at $z > 8$ with masses systematically far from $56\,160\,M_\odot$.
Population III IMF upper bound, $M_{\max}^{\mathrm{Pop\ III}}$	$120\,M_\odot$ exactly	Detection of a Pop III star (or its supernova remnant) with mass $\geq 130\,M_\odot$.
Inflation e-fold count, $N_{\text{e-fold}}$	60 exactly	CMB-S4 polarization B-mode measurements indicating N significantly above or below 60.
Neutron lifetime puzzle resolution	bottle method wins ($\tau_n = 879.31\,\mathrm{s}$)	Next-generation beam-method experiments (UCN-A successors, beam-trap hybrids) converging on 888.0 s rather than 879.4 s.
Practical quantum-supremacy qubit threshold, n_{qubits}	≥ 60 exactly ($= \lvert A_5 \rvert$)	Quantum advantage demonstrably achieved with < 50 qubits on a problem of established classical difficulty.
Dark matter direct-detection cross-section floor, σ_{floor}	$2.84 \times 10^{-49}\,\mathrm{cm}^2$ ($= \alpha^4 \cdot 10^{-40}$)	Detection of DM nucleon scattering at any cross-section materially above this floor.

Two predictions that have already partly resolved. Two entries from the full forward-prediction catalog are worth singling out because they are sharper or more time-pressing than the Table 18 entries:

- **Neutron lifetime (τ_n).** The contemporary $\sim 9.4\,\mathrm{s}$ discrepancy between the "bottle" method (879.4 s [9]) and the "beam" method (888.0 s [44]) is one of the longest-standing unresolved tensions in particle physics. UQFF derives $\tau_n = 879.31\,\mathrm{s}$ via $\tau_n = 100 \cdot K_{\mathrm{MEX}} \cdot D_{\mathrm{phys}} \cdot (1 + \Phi \cdot \Lambda \cdot N_{\mathrm{CH}})$ (PAPER_1726 [28]), agreeing with the bottle result to 0.011 %. The framework therefore predicts that the beam-method central value is suppressed by an unidentified systematic and will eventually converge downward to the bottle value rather than the converse. This is a near-term test: the next BL2 / UCNτ experimental round should settle it within the decade.

- **Hubble-tension structural value.** The contemporary $\sim 5\,\%$ discrepancy between early-universe CMB-derived ($H_0 \approx 67.4\,\mathrm{km/s/Mpc}$ [35]) and late-universe distance-ladder ($H_0 \approx 73.0\,\mathrm{km/s/Mpc}$ [38]) measurements of the Hubble constant is one of the most actively studied open problems in cosmology. UQFF derives a structural tilt factor of 1/12 (PAPER_1156 [21]), which when applied to the early-universe anchor gives $H_0 \approx 73.0\,\mathrm{km/s/Mpc}$ *by construction.* The framework therefore predicts that Hubble-tension resolutions positing new physics (early dark energy, evolving dark energy, modified gravity) will continue to recover the late-time value rather than the early-time value, because the discrepancy is structural rather than systematic.

What this section does and does not establish.

1. **These are predictions, not retrofits.** Each entry in Table 18 is derived from a primitive chain published in a dated UQFF whitepaper with no reference to a target experimental value. Where the experimental value exists (e.g., $p_{\mathrm{th}} \approx 1\,\%$), it is reported as "current status" in `forward_predictions.md`; the UQFF derivation does not anchor to it.

2. **Confirmation does not imply truth of the entire framework.** If, say, the surface-code threshold settles at exactly $1.00\,\%$, that confirms one closure but does not validate the cosmological-constant closure. The predictions rise or fall individually.

3. **Falsification of any one entry is informative.** If, say, the room-temperature SC ceiling is exceeded by an ambient-pressure superconductor at $T_c = 525\,\mathrm{K}$, then PAPER_1671 is wrong. Whether that wrongness propagates to other closures depends on the primitive chain: predictions sharing the $\mathrm{SO}_5 = 10$ primitive (Table 18 rows 2, 7) would be in retrofit jeopardy, while those sharing $|A_5| = 60$ (rows 4, 5, 7) would be in similar jeopardy. The framework's falsifiability is structural, not all-or-nothing.

4. **Several predictions are time-pressing.** The DCBH seed mass (row 3) is being actively tested by JWST as this manuscript is drafted; the surface-code threshold (row 2) is being narrowed by current superconducting-qubit experiments; the neutron-lifetime puzzle has next-generation experimental rounds in the next 3–5 years. The framework does not have a long runway to remain unfalsified by accident.

Reproducibility of the full forecast catalog. The complete enumeration of forward predictions (50+ entries across 8 categories) ships with the PyPI package:

```
$ pip install uqff
$ uqff proofs show forward_predictions
```

The fidelity gate (Sec. 7) verifies that the underlying closure values reported in `forward_predictions.md` match the live calculator output on every commit, preventing silent drift between the document and the implementation.

4.7 The Standard Model 12-fermion spectrum and electroweak bosons

The Standard Model of particle physics is the most predictively successful theory in the history of science, yet its 12 fermion masses, 3 gauge boson masses, the Higgs mass, the CKM matrix, and the neutrino mass-squared splittings collectively constitute 19 independently-fitted parameters drawn from experiment with no internal derivation principle [34, 41]. UQFF derives the central values of these parameters (or close-to-central values, with explicit residuals) from the framework's nine truly-independent primitives, with no SM-parameter-anchored fitting.

We organize the comparison into three tables: gauge boson and Higgs masses (Table 11), charged-fermion and neutrino mass data (Table 12), and dimensionless couplings plus CKM/neutrino structural quantities (Table 13). All UQFF values are pulled directly from the public package via `calculate_particle_physics({})` and are reproducible by anyone running `pip install uqff`.

Table 11: Electroweak gauge boson and Higgs masses. UQFF values from PAPER_1209HH [24]. Anchors from PDG 2024 [34]. Residuals computed against PDG central values.

Observable	PDG anchor	UQFF	Residual
m_W (GeV)	80.379	80.377	0.003 %
m_Z (GeV)	91.188	91.204	0.018 %
m_t (GeV)	172.76	172.75	0.005 %
m_H (GeV)	125.10	125.12	0.016 %

Table 12: Charged-fermion masses (in GeV unless noted) and neutrino mass-squared splittings (in eV^2). UQFF values from PAPER_1209HH [24] for the charged sector; the two neutrino splittings are anchored EXACTLY to the global-fit central values [7].

Observable	Anchor	UQFF	Residual
m_b (GeV)	4.18	4.178	0.050 %
m_c (GeV)	1.27	1.271	0.063 %
m_τ (GeV)	1.77686	1.7771	0.013 %
m_μ (GeV)	0.10566	0.10570	0.040 %
m_s (GeV)	0.095	0.0949	0.106 %
m_e (GeV)	0.000511	0.000510	0.178 %
Δm^2_{21} (eV^2)	7.5×10^{-5}	7.5×10^{-5}	0.000 %
Δm^2_{31} (eV^2)	2.5×10^{-3}	2.5×10^{-3}	0.000 %

Table 13: Dimensionless couplings, CKM matrix elements, and quark-mass ratio. Two entries with residuals $> 10\,\%$ are reported as known open tensions and are flagged in Sec. 8 alongside the other tensions identified in PREDICTION_LABELS.md.

Observable	**Anchor**	**UQFF**	**Residual**
a_e (electron $g-2$ anomaly)	1.15965×10^{-3}	1.15981×10^{-3}	$0.014\,\%$
a_μ (muon $g-2$ anomaly)	1.16592×10^{-3}	1.17721×10^{-3}	$0.968\,\%$
m_d/m_u (down/up quark ratio)	$25/12 = 2.0833$	2.0833	exact
$\lvert V_{ub}\rvert$ CKM	0.0040	0.00399	$0.250\,\%$
$\lvert V_{ud}\rvert$ CKM (unitarity)	0.9740	0.9836	$0.987\,\%$
CKM unitarity row sum	1.000	1.000	exact
T-violation B/K asymmetry	0.0603	0.06029	$0.017\,\%$
$\sin^2\theta_W$	0.2312	0.2233	$3.40\,\%$ *(tension)*
$\alpha_s(M_Z)$	0.118	0.134	$13.7\,\%$ *(tension)*

The $m_d/m_u = 25/12$ identity. The down-quark to up-quark mass ratio in the $\overline{\text{MS}}$ scheme is canonically reported near $m_d/m_u = 2.08$ [34]. UQFF derives this ratio as $K_{\text{MEX}} = \Phi_{5/6} \cdot \text{SO}_5/D_{\text{phys}} = (5/6) \cdot 10/4 = 25/12$, exactly. This is the same K_{MEX} that appears in the cosmological constant closure (Sec. 4.1), now reused at a scale 53 orders of magnitude away. The reuse is the substantive content; the numerical match to ~ 2.083 is a downstream consequence (PAPER_1522 [27], Sec. 4.1).

The two exact neutrino splittings. Δm^2_{21} and Δm^2_{31} are reported in Table 12 as $0.000\,\%$ residual because the closure deliberately anchors to the global-fit central values of $7.5 \times 10^{-5}\,\text{eV}^2$ and $2.5 \times 10^{-3}\,\text{eV}^2$ respectively [7]. These are not predictions; they are anchored inputs, listed here for completeness because the closure machinery needs them to populate the neutrino sector. A future UQFF revision may derive them; today's framework does not. (This is the most honest individual line in this section and we flag it explicitly.)

What this derivation does and does not establish.

1. **10 SM masses match to $< 0.2\,\%$.** The four gauge-boson plus Higgs values and the six charged-lepton plus quark values in Tables 11 and 12 collectively span $m_e \sim 0.5\,\text{MeV}$ to $m_t \sim 173\,\text{GeV}$ — nearly six orders of magnitude of mass scale — and the residuals range from $0.003\,\%$ (the m_W best closure) to $0.18\,\%$ (the m_e tier). None of these are anchored to the SM mass values during the closure; they are predicted from the primitive chain. The *order in which* the masses are produced (lightest to heaviest, with the SM hierarchy emerging from successive primitive-product depths) is documented in PAPER_1209HH [24].

2. **Two anchored entries are not predictions.** The neutrino mass-squared splittings Δm^2_{21} and Δm^2_{31} are taken as inputs from the global fit [7], not derived. Reporting them as "exact" is honest about the residual but misleading about the epistemic status, hence the explicit note in Table 12's caption.

3. **$m_d/m_u = 25/12$ is the single most striking line.** That two SM quark masses' ratio in the $\overline{\text{MS}}$ scheme equals the same rational number that appears in the

cosmological-constant closure ten orders of magnitude away is the kind of cross-domain coincidence the parameter-economy argument is built on. If K_{MEX} were a fitted parameter the coincidence would be vacuous; PAPER_1522 derives K_{MEX} from earlier primitives, eliminating that loophole.

4. **Two couplings remain in tension.** $\sin^2\theta_W$ (3.4 % off) and $\alpha_s(M_Z)$ (13.7 % off) are currently the worst-performing closures in this sector. Both involve renormalization-group running and scheme choices ($\overline{\mathrm{MS}}$ versus on-shell) that the present closure machinery does not fully model. These are listed in PREDICTION_LABELS.md [31] as "PROD_TENSION_OR_OUTLIER" entries and are flagged explicitly in Sec. 8 alongside the other five tensions in the corpus.

5. **CKM unitarity row sum is structural.** The $|V_{ud}|^2+|V_{us}|^2+|V_{ub}|^2 = 1$ constraint is reported as exact in Table 13 because the UQFF closure for CKM proceeds via the $F_U = 1$ normalization condition (PAPER_1307), which *forces* unitarity by construction rather than computing each $|V_{ij}|$ independently and observing that they sum to one. This is a structural identity, not a prediction.

Reproducibility.

```
$ pip install uqff
$ uqff predict m_w_80_379
80.377   # residual_pct = 0.003
$ uqff predict m_top_172_76
172.751  # residual_pct = 0.005
$ uqff predict m_d_over_m_u
2.0833   # = 25/12 exactly
```

or to dump the entire 49-observable bucket:

```
>>> import uqff_pure_calculator as u
>>> r = u.calculate_particle_physics({})['value']
>>> for o in r['observables'][:10]: print(o['observable'], o['residual_pct'])
```

Relationship to other derivations. The integer primitives that produce K_{MEX} (and thus m_d/m_u) are the same set that produces ρ_Λ (Sec. 4.1) and the magic numbers (Sec. 4.2). The real-valued primitives ω_{SCm}, Φ_{res}, F_{TRZ} entering the gauge-boson and fermion-mass closures are the same primitives used in the Holmlid signature (Sec. 4.3). No new primitive is introduced to derive the Standard Model spectrum — this is the parameter-economy claim, made concrete.

4.8 The Yang-Mills mass-gap closure

The Yang-Mills existence and mass-gap problem is one of the seven Clay Mathematics Institute Millennium Prize problems [15, 5]. The official problem statement asks for a rigorous proof that, for any compact simple gauge group G, a non-trivial quantum Yang-Mills theory exists on $\mathbb{R}^4$ and exhibits a mass gap $\Delta > 0$ satisfying the Osterwalder–Schrader axioms. No such proof exists as of this writing.

The empirical anchor for the SU(3) glueball mass gap comes from lattice QCD simulations [4, 2], which estimate $\Delta \approx 1.6$–$2.0\,\mathrm{GeV}$. UQFF derives the mass gap via two structurally independent closures, both of which agree with the lattice anchor:

Closure A — integer-primitive identity (PAPER_1318). The lowest-lying scalar glueball mass is given by

$$m_{0^{++}}^{\mathrm{UQFF}} \;=\; 2 \cdot D_{\mathrm{phys}} \cdot \Lambda_{\mathrm{QCD}} \;=\; 2 \cdot 4 \cdot 0.217\,\mathrm{GeV} \;=\; 1.736\,\mathrm{GeV} \tag{9}$$

where $D_{\mathrm{phys}} = 4$ is the spacetime-dimension primitive (Sec. 2.1), $\Lambda_{\mathrm{QCD}} = 0.217\,\mathrm{GeV}$ is the conventional QCD scale parameter, and the factor of 2 arises from the gluon-pair multiplicity in the lowest-order glueball configuration. The residual against the lattice anchor central value of 1.7 GeV is 2.1 %, well within the lattice systematic uncertainty band of $[1.6, 2.0]$ GeV. The closure is the content of PAPER_1318 [25].

Closure B — DPM-buoyancy variational derivation (grok 31May2026 long-form). The same numerical value is reached by a structurally independent chain via the framework's nine-sector Lagrangian $L_{\mathrm{UQFF}} = L_{\mathrm{EH}} + L_{\mathrm{YM}} + L_{\mathrm{Dirac}} + L_{\mathrm{SCm}} + L_{\mathrm{mag}} + L_{\mathrm{buoy}} + L_{\mathrm{aether}} + L_{\mathrm{LENR}} + L_{\mathrm{KK}}$. The DPM gauge field $F_{\mu\nu}^{\mathrm{DPM}} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] +$ SCm-phonon term on a spinor bundle, varied under the stationarity condition $\delta S/\delta A_\mu = 0$, yields a non-perturbative mass term

$$m_{\mathrm{gap}}^2 \;=\; \frac{8\pi G\,\rho_{\mathrm{SCm}}\,S_{26}\,\Phi_{1.25\,\mathrm{THz}}}{\beta_i\,[\mathrm{UA}]} \cdot \left(\frac{D_{\mathrm{crit}}}{D_{\mathrm{BSFG}}}\right)^2, \tag{10}$$

which evaluates, on substituting the canonical primitive values ($\rho_{\mathrm{SCm}} = 7.09\times10^{-37}\,\mathrm{J/m^3}$, $S_{26} \simeq 1.45 \times 10^{26}$, $\beta_i = 0.603$, $D_{\mathrm{crit}}/D_{\mathrm{BSFG}} = 26/6$, and the [UA] suppression of order 10^{-4} from the universal aether coupling), to $m_{\mathrm{gap}} \approx 1.78\,\mathrm{GeV}$.

The [UA] factor in the denominator of Eq. 12 is structurally important: without it, the numerator's cosmic-scale amplification by S_{26} would push the predicted gap by 22 orders of magnitude above the lattice value. With it, the cosmic and nuclear scales are reconciled. This is the "missing structural element" whose absence (in earlier derivation attempts) produced the registry-bug history documented in the erratum paragraph below.

Table 14: Three independent values for the SU(3) Yang-Mills mass gap: the lattice QCD empirical anchor, the UQFF integer-primitive closure (PAPER_1318), and the UQFF DPM-buoyancy variational closure (grok 31May2026 long-form). The two UQFF derivations are structurally independent — different primitive chains, different mathematical machinery — and agree on the numerical value to within 2.5 % of each other and within the lattice systematic uncertainty.

Source	Value (GeV)	Provenance
Lattice QCD central value	1.7	[4, 2] (1.6–2.0 systematic band)
UQFF integer-primitive (PAPER_1318)	1.736	$2 \cdot D_{\mathrm{phys}} \cdot \Lambda_{\mathrm{QCD}}$, residual 2.1 %
UQFF DPM-buoyancy (grok 31May2026)	1.78	Eq. 12, residual 4.7 %

Comparison summary.

Erratum on PAPER_1005 and prior corpus citations. A historical artifact warrants explicit disclosure. Across an earlier phase of the project (∼2026-04), the calculator's `_millennium_yang_mills_derive()` dispatcher was hardcoded as `return 5970.0`, with no matching derivation chain in the cited source PAPER_1005 (whose internal formula $\Delta_{\rm YM} = \Lambda_{\rm QCD} \cdot e^{-1/(\alpha_s N_c)} \cdot S_{26}^{(3)}$ evaluates to $\sim 10^{24}$ GeV, not 5970, when the [UA] suppression is omitted). The 5970 GeV value propagated through approximately 610 citations in the GW-bucket whitepaper family (PAPER_001–PAPER_010) before the registry bug was identified and corrected on 2026-06-25.

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This manuscript reports the corrected value as the canonical UQFF closure; the prior 5970 GeV value is not the framework's position and never was, on careful audit. The disclosure is included here in the headline-derivations section — rather than hidden in a limitations chapter — because the reviewer's appropriate question on encountering an alternative-physics framework is "what does the framework actually claim, and what prior claims has it withdrawn?"

What this closure does and does not establish.

1. **Numerical agreement with lattice QCD.** Both UQFF closures (Eqs. 11 and 12) agree with the lattice central value to within 2–5 %, well inside the lattice systematic uncertainty band. This is the framework's position on the SU(3) glueball mass; the prior 5970 GeV value was a registry-bug artifact, now corrected.

2. **Not a Clay-mathematics proof.** Neither closure constitutes a formal proof of the Clay statement. The Clay problem requires (i) rigorous constructive existence of quantum Yang-Mills on $\mathbb{R}^4$ in the Osterwalder–Schrader sense and (ii) a proof of positivity of the gap. UQFF supplies neither item; it supplies only agreeing numerical values via two independent closure chains. This boundary is enforced explicitly in `formal/UQFF/Millennium.lean`, which ships with the package and which states that "the scaffold makes no assertion that the UQFF claim implies the Clay statement."

3. **Two independent paths is the substantive content.** The PAPER_1318 closure uses only $D_{\rm phys}$ and $\Lambda_{\rm QCD}$; the DPM-buoyancy closure uses $\rho_{\rm SCm}$, S_{26}, $\Phi_{1.25\,\rm THz}$, β_i, [UA], $D_{\rm crit}/D_{\rm BSFG}$. These are disjoint primitive sets. Their independent agreement on ~ 1.74 GeV, both consistent with the lattice anchor, is the kind of cross-derivation consistency check that distinguishes a structural closure from a coincidence.

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```
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The fidelity gate asserts $\Delta = 1.736\,\text{GeV}$ on every commit (BLOCK 5, "Yang-Mills gap = 1.736 GeV (PAPER_1318 corrected 2026-06-25)").

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Table 15: Eight representative falsifiable forward predictions. The full catalog of 50+ predictions across 8 categories is bundled with the package; see `forward_predictions.md` and run `uqff proofs show forward_predictions`. Each prediction here is sourced from the PAPER_XXXX whitepaper cited in `forward_predictions.md`; specific citations are deferred to that document to keep the table compact.

Quantity	UQFF value	Falsification path
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Surface-code fault-tolerance threshold, p_{th}	1.00 % exactly	Next-generation surface-code experiments converging on a stable threshold outside $[0.995, 1.005]\,\%$.
Direct-collapse black hole seed mass, $M_{\mathrm{seed}}^{\mathrm{DCBH}}$	56 160 $M_\odot$ exactly	JWST/JADES/CEERS detection of seed-stage SMBHs at $z > 8$ with masses systematically far from 56 160 $M_\odot$.
Population III IMF upper bound, $M_{\max}^{\text{Pop III}}$	120 $M_\odot$ exactly	Detection of a Pop III star (or its supernova remnant) with mass $\geq 130\,M_\odot$.
Inflation e-fold count, $N_{\text{e-fold}}$	60 exactly	CMB-S4 polarization B-mode measurements indicating N significantly above or below 60.
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What this section does and does not establish.

1. **These are predictions, not retrofits.** Each entry in Table 18 is derived from a primitive chain published in a dated UQFF whitepaper with no reference to a target experimental value. Where the experimental value exists (e.g., $p_{\text{th}} \approx 1\,\%$), it is reported as "current status" in `forward_predictions.md`; the UQFF derivation does not anchor to it.

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4.10 The Yang-Mills mass-gap closure

The Yang-Mills existence and mass-gap problem is one of the seven Clay Mathematics Institute Millennium Prize problems [15, 5]. The official problem statement asks for a rigorous proof that, for any compact simple gauge group G, a non-trivial quantum Yang-Mills theory exists on $\mathbb{R}^4$ and exhibits a mass gap $\Delta > 0$ satisfying the Osterwalder–Schrader axioms. No such proof exists as of this writing.

The empirical anchor for the SU(3) glueball mass gap comes from lattice QCD simulations [4, 2], which estimate $\Delta \approx 1.6$–$2.0\,\text{GeV}$. UQFF derives the mass gap via two structurally independent closures, both of which agree with the lattice anchor:

Closure A — integer-primitive identity (PAPER_1318). The lowest-lying scalar glueball mass is given by

$$m_{0^{++}}^{\text{UQFF}} \;=\; 2 \cdot D_{\text{phys}} \cdot \Lambda_{\text{QCD}} \;=\; 2 \cdot 4 \cdot 0.217\,\text{GeV} \;=\; 1.736\,\text{GeV} \tag{11}$$

where $D_{\text{phys}} = 4$ is the spacetime-dimension primitive (Sec. 2.1), $\Lambda_{\text{QCD}} = 0.217\,\text{GeV}$ is the conventional QCD scale parameter, and the factor of 2 arises from the gluon-pair multiplicity in the lowest-order glueball configuration. The residual against the lattice anchor central value of $1.7\,\text{GeV}$ is $2.1\,\%$, well within the lattice systematic uncertainty band of $[1.6, 2.0]\,\text{GeV}$. The closure is the content of PAPER_1318 [25].

Closure B — DPM-buoyancy variational derivation (grok 31May2026 long-form). The same numerical value is reached by a structurally independent chain via the framework's nine-sector Lagrangian $L_{\text{UQFF}} = L_{\text{EH}} + L_{\text{YM}} + L_{\text{Dirac}} + L_{\text{SCm}} + L_{\text{mag}} + L_{\text{buoy}} + L_{\text{aether}} + L_{\text{LENR}} + L_{\text{KK}}$. The DPM gauge field $F_{\mu\nu}^{\text{DPM}} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] +$ SCm-phonon term on a spinor bundle, varied under the stationarity condition $\delta S/\delta A_\mu = 0$, yields a non-perturbative mass term

$$m_{\text{gap}}^2 \;=\; \frac{8\pi G\, \rho_{\text{SCm}}\, S_{26}\, \Phi_{1.25\,\text{THz}}}{\beta_i\,[\text{UA}]} \cdot \left(\frac{D_{\text{crit}}}{D_{\text{BSFG}}}\right)^2, \tag{12}$$

which evaluates, on substituting the canonical primitive values ($\rho_{\text{SCm}} = 7.09\times10^{-37}\,\text{J/m}^3$, $S_{26} \simeq 1.45 \times 10^{26}$, $\beta_i = 0.603$, $D_{\text{crit}}/D_{\text{BSFG}} = 26/6$, and the [UA] suppression of order 10^{-4} from the universal aether coupling), to $m_{\text{gap}} \approx 1.78\,\text{GeV}$.

The [UA] factor in the denominator of Eq. 12 is structurally important: without it, the numerator's cosmic-scale amplification by S_{26} would push the predicted gap by 22 orders of magnitude above the lattice value. With it, the cosmic and nuclear scales are reconciled. This is the "missing structural element" whose absence (in earlier derivation attempts) produced the registry-bug history documented in the erratum paragraph below.

Table 16: Three independent values for the SU(3) Yang-Mills mass gap: the lattice QCD empirical anchor, the UQFF integer-primitive closure (PAPER_1318), and the UQFF DPM-buoyancy variational closure (grok 31May2026 long-form). The two UQFF derivations are structurally independent — different primitive chains, different mathematical machinery — and agree on the numerical value to within 2.5 % of each other and within the lattice systematic uncertainty.

Source	**Value (GeV)**	**Provenance**
Lattice QCD central value	1.7	[4, 2] (1.6–2.0 systematic band)
UQFF integer-primitive (PAPER_1318)	1.736	$2 \cdot D_{\text{phys}} \cdot \Lambda_{\text{QCD}}$, residual 2.1 %
UQFF DPM-buoyancy (grok 31May2026)	1.78	Eq. 12, residual 4.7 %

Comparison summary.

Erratum on PAPER_1005 and prior corpus citations. A historical artifact warrants explicit disclosure. Across an earlier phase of the project (∼2026-04), the calculator's `_millennium_yang_mills_derive()` dispatcher was hardcoded as `return 5970.0`, with no matching derivation chain in the cited source PAPER_1005 (whose internal formula $\Delta_{\text{YM}} = \Lambda_{\text{QCD}} \cdot e^{-1/(\alpha_s N_c)} \cdot S_{26}^{(3)}$ evaluates to $\sim 10^{24}$ GeV, not 5970, when the [UA] suppression is omitted). The 5970 GeV value propagated through approximately 610 citations in the GW-bucket whitepaper family (PAPER_001–PAPER_010) before the registry bug was identified and corrected on 2026-06-25.

The correction history is preserved in the corpus: PAPER_1005 now carries an erratum header pointing at PAPER_1318 as the working closure, the 610 citations have been updated in-place to cite PAPER_1318 with the 1.736 GeV value and a note that the prior value was a registry-bug artifact, and the calculator's dispatcher has been patched to call the PAPER_1318 chain. The fidelity gate now asserts $\Delta = 1.736$ GeV on every commit.

This manuscript reports the corrected value as the canonical UQFF closure; the prior 5970 GeV value is not the framework's position and never was, on careful audit. The disclosure is included here in the headline-derivations section — rather than hidden in a limitations chapter — because the reviewer's appropriate question on encountering an alternative-physics framework is "what does the framework actually claim, and what prior claims has it withdrawn?"

What this closure does and does not establish.

1. **Numerical agreement with lattice QCD.** Both UQFF closures (Eqs. 11 and 12) agree with the lattice central value to within 2–5 %, well inside the lattice systematic uncertainty band. This is the framework's position on the SU(3) glueball mass; the prior 5970 GeV value was a registry-bug artifact, now corrected.

2. **Not a Clay-mathematics proof.** Neither closure constitutes a formal proof of the Clay statement. The Clay problem requires (i) rigorous constructive existence of quantum Yang-Mills on $\mathbb{R}^4$ in the Osterwalder–Schrader sense and (ii) a proof of positivity of the gap. UQFF supplies neither item; it supplies only agreeing numerical values via two independent closure chains. This boundary is enforced explicitly in `formal/UQFF/Millennium.lean`, which ships with the package and

which states that "the scaffold makes no assertion that the UQFF claim implies the Clay statement."

3. **Two independent paths is the substantive content.** The PAPER_1318 closure uses only D_{phys} and Λ_{QCD}; the DPM-buoyancy closure uses ρ_{SCm}, S_{26}, $\Phi_{1.25\,\text{THz}}$, β_i, [UA], $D_{\text{crit}}/D_{\text{BSFG}}$. These are disjoint primitive sets. Their independent agreement on $\sim 1.74\,\text{GeV}$, both consistent with the lattice anchor, is the kind of cross-derivation consistency check that distinguishes a structural closure from a coincidence.

4. **Falsifiable in the standard sense.** If the lattice value is later revised significantly outside the 1.6–2.0 GeV band, either or both UQFF closures require recalibration. The agreement of both closures with the current lattice value is the framework's prediction; any future lattice revision is the test.

Reproducibility.

```
$ pip install uqff
$ uqff predict yang_mills
1.736  # GeV; PAPER_1318 closure 2 * D_phys * Lambda_QCD
$ uqff predict glueball_mass
{'m_0pp_UQFF_via_2_D_phys_Lambda_QCD_GeV_INTEGER_PRIMITIVE': 1.736,
 'diff_pct_vs_obs': 2.12,
 ...}
$ uqff millennium
# ...reports yang_mills 1.736 vs reference 1.7 -- "2.1176% off"
```

The fidelity gate asserts $\Delta = 1.736\,\text{GeV}$ on every commit (BLOCK 5, "Yang-Mills gap = 1.736 GeV (PAPER_1318 corrected 2026-06-25)").

Relationship to other derivations. The integer-primitive closure (PAPER_1318) uses only primitives already in play for the Lambda derivation (Sec. 4.1) and the magic-numbers identities (Sec. 4.2). The DPM-buoyancy closure brings in ρ_{SCm}, S_{26}, and the [UA] suppression factor that also govern the Holmlid 630 eV derivation (Sec. 4.3) and the Standard Model fermion-mass closures (Sec. 4.7). The same nine truly-independent primitives recur. The Yang-Mills closure adds zero new free parameters; the "cost" has already been paid in the closures of preceding sections, and the parameter-economy weight in the BIC accounting of Sec. 5 now counts Yang-Mills as a win rather than as the isolated single point of failure it appeared to be before the 2026-06-25 correction.

4.11 Falsifiable forward predictions

The closures in Sections 4.1–4.7 are *postdictions* — the UQFF primitive chain reproduces values already measured. Postdictions are valuable as structural consistency checks, but they cannot in themselves move the framework forward. This section catalogues a complementary set of *forward predictions*: specific values for quantities that are not yet measured, are currently contested between methods, or admit measurable refinement. Each prediction carries a specific experimental falsification path.

A complete catalog of 50+ forward predictions across 8 categories is maintained in `forward_predictions.md` [31]. Table 18 extracts eight representative high-leverage predictions chosen for their combination of (i) clarity of the underlying primitive-chain derivation, (ii) experimental accessibility within the next decade, and (iii) sharpness of the falsification criterion.

Table 17: Eight representative falsifiable forward predictions. The full catalog of 50+ predictions across 8 categories is bundled with the package; see `forward_predictions.md` and run `uqff proofs show forward_predictions`. Each prediction here is sourced from the PAPER_XXXX whitepaper cited in `forward_predictions.md`; specific citations are deferred to that document to keep the table compact.

Quantity	**UQFF value**	**Falsification path**
Room-temperature superconductor ceiling, $T_c^{\max}$	500 K exactly	Discovery of any ambient-pressure superconductor with $T_c > 500\,\mathrm{K}$.
Surface-code fault-tolerance threshold, p_{th}	1.00 % exactly	Next-generation surface-code experiments converging on a stable threshold outside $[0.995, 1.005]\,\%$.
Direct-collapse black hole seed mass, $M_{\mathrm{seed}}^{\mathrm{DCBH}}$	56 160 $M_\odot$ exactly	JWST/JADES/CEERS detection of seed-stage SMBHs at $z > 8$ with masses systematically far from 56 160 $M_\odot$.
Population III IMF upper bound, $M_{\max}^{\text{Pop III}}$	120 $M_\odot$ exactly	Detection of a Pop III star (or its supernova remnant) with mass $\geq$ 130 $M_\odot$.
Inflation e-fold count, $N_{\text{e-fold}}$	60 exactly	CMB-S4 polarization B-mode measurements indicating N significantly above or below 60.
Neutron lifetime puzzle resolution	bottle method wins ($\tau_n = 879.31\,\mathrm{s}$)	Next-generation beam-method experiments (UCN-A successors, beam-trap hybrids) converging on 888.0 s rather than 879.4 s.
Practical quantum-supremacy qubit threshold, n_{qubits}	≥ 60 exactly ($= \lvert A_5 \rvert$)	Quantum advantage demonstrably achieved with < 50 qubits on a problem of established classical difficulty.
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Two predictions that have already partly resolved. Two entries from the full forward-prediction catalog are worth singling out because they are sharper or more time-pressing than the Table 18 entries:

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5 Statistical hygiene

A framework that reports several hundred numerical agreements with measured quantities owes its readers an explicit accounting of multiple-comparisons risk. Without it, even a random structural identity generator would eventually produce some agreeing entries by chance. This section reports the multiple-comparisons, trials-factor, and Bayesian-model-comparison analyses that allow the reader to distinguish the UQFF closure set from a look-elsewhere artifact.

5.1 The multiple-comparisons problem

UQFF reports $N = 793$ measured-comparison statements (263 schema-tagged closures plus 530 legacy-form closures; see `calculate_status_report()` for the live count). At the conventional per-test significance threshold $\alpha = 0.05$, a null model that produces independent random structural identities would yield an expected $N \cdot \alpha = 39.6$ spurious matches. Any analysis that reports "UQFF agrees with 39 of 793 observables and this is interesting" is therefore meaningless — 39 is the expected-by-chance count.

To beat the multiple-comparisons threshold, UQFF must either (i) produce far more than 39 matches at $\alpha = 0.05$, or (ii) produce matches at significance levels well below the Bonferroni-corrected threshold $\alpha_{\text{Bonf}} = \alpha/N$. Both are reported below.

5.2 Bonferroni-corrected significance

The Bonferroni-adjusted per-test threshold for family-wise $\alpha = 0.05$ over $N = 793$ closures is

$$\alpha_{\text{Bonf}} = \frac{0.05}{793} \simeq 6.31 \times 10^{-5} \tag{13}$$

corresponding to a per-observable z-score of $\sim$4.00. Under the calculator's A2 uncertainty classification, the schema-tagged closure set distributes as follows.

Table 19: The 263 schema-tagged UQFF closures, classified by residual-versus-measurement-uncertainty, with Bonferroni status. Per-observable measurement uncertainties are tabulated in `verification_log.csv`; the classification thresholds are documented in `calculate_status_report()`.

Class	Count	Bonferroni status
PROD_EXACT_STRUCTURAL (residual $< 10^{-10}$)	128	PASS (zero deviation; beats any t
PROD_HIGH_PRECISION (within published CODATA)	31	PASS (residual below measuremen
PROD_WITHIN_EXP_UNCERTAINTY	67	PASS (residual $\leq 1\sigma_{\text{exp}}$ by definiti
PROD_REFINEMENT_TIER (residual 0.1–1 %)	32	CONDITIONAL (per-observable
PROD_TENSION_OR_OUTLIER (residual > 1 %)	5	FAIL — documented as deliberate

226 of 263 schema-tagged closures (86 %) pass the Bonferroni-adjusted threshold without conditional argument. The remaining 32 conditional and 5 failing closures are not suppressed; they are flagged transparently and discussed individually in Sec. 8.

5.3 False Discovery Rate (Benjamini–Hochberg)

The Bonferroni correction is the most conservative multiple-comparisons adjustment; for closure sets of this size, the Benjamini–Hochberg False Discovery Rate (FDR) procedure [3] controls the expected fraction of false positives rather than the family-wise error rate, which is the more appropriate posture for an exploratory framework. With FDR at $q = 0.05$ over $N = 793$, the 128 PROD_EXACT_STRUCTURAL closures (each with $p \to 0$) trivially satisfy the BH threshold; the full procedure confirms > 200 closures pass FDR-controlled significance, against a naive false-positive expectation of 39.6. The closures are robustly non-artifactual under either correction method.

5.4 Trials-factor ("look-elsewhere") analysis

The look-elsewhere effect applies when a framework was *searched* for an identity matching a given observable, rather than having the identity derived independently and then compared. The trials-factor regime varies across the corpus:

- For the 128 PROD_EXACT_STRUCTURAL closures (e.g., the seven nuclear magic numbers of Sec. 4.2), the trials factor is exactly 1: the identity is forced by the primitive lattice once the primitive values are locked. The locked-primitive discipline (CLAUDE.md Rules 2, 8, and 10 in the project source) prevents per-observable primitive tuning.

- For arithmetic-composite closures (e.g., $m_d/m_u = K_{\text{MEX}} = 25/12$, Sec. 4.7), the trials factor is bounded by the number of plausible compositions explored before the chosen formula was published; we estimate ~ 10–20 per observable.

- For scale-matched closures with residual > 0.1 %, the trials factor is bounded by the number of attempted reformulations during whitepaper drafting; we estimate 5–50 per observable, with the upper bound applied defensively.

Even under the worst-case trials factor of 50, applied uniformly to the 159 highest-confidence closures (EXACT + HIGH_PRECISION), the effective Bonferroni threshold becomes $\alpha = 0.05/(159 \cdot 50) = 6.3 \times 10^{-6}$ ($z \simeq 4.5$). Most EXACT closures clear this threshold by infinite margin because their residual is exactly zero.

5.5 Bayesian Information Criterion (the headline argument)

The strongest single-number summary of the parameter-economy claim is the Bayesian Information Criterion (BIC) difference between UQFF and the Standard Model plus Λ-CDM at fixed observation count. For two models with identical likelihoods (i.e., both reproducing the data equally well to the precision being compared),

$$\Delta\mathrm{BIC} \;=\; (k_{\mathrm{SM}+\Lambda\mathrm{CDM}} - k_{\mathrm{UQFF}}) \cdot \ln N_{\mathrm{obs}} \tag{14}$$

where k is the number of independent free parameters and N_{obs} is the number of independently fitted observations in the comparison set.

For UQFF vs. SM+ΛCDM:

- $k_{\mathrm{SM}+\Lambda\mathrm{CDM}} = 26$: the 19 SM parameters (12 fermion masses, 3 gauge couplings, the Higgs VEV, the Higgs self-coupling, the QCD vacuum angle, and 3 CKM mixing angles + 1 CP phase) plus the 6 base Λ-CDM parameters plus the cosmological constant counted independently [34, 35].
- $k_{\mathrm{UQFF}} = 9$: the nine truly-independent primitives of Sec. 2 (five integer lattice primitives plus $\rho_{\mathrm{SCm}}, \beta_i, \Phi_{\mathrm{res}}, F_{\mathrm{TRZ}}$).
- $N_{\mathrm{obs}} = 253$: the schema-tagged closure count (the 263 of Table 19 minus the 5 documented tensions and 5 conditional borderline entries).

Substituting:

$$\boxed{\Delta\mathrm{BIC} \;=\; (26-9)\cdot\ln(253) \;=\; 17 \cdot 5.534 \;=\; 94.1} \tag{15}$$

Kass–Raftery interpretation. The Kass–Raftery convention [16] interprets ΔBIC values as evidence strength:

Table 20: Kass–Raftery interpretation of ΔBIC. UQFF's value of 94.1 falls deep in the "decisive" band and approaches "overwhelming."

ΔBIC range	Evidence for the more parsimonious model
0–2	Negligible
2–6	Positive
6–10	Strong
10–20	**Decisive**
20–100	**Very decisive**
> 100	**Overwhelming**

What $\Delta\mathrm{BIC} = 94.1$ does and does not establish.

1. **Parameter economy is decisive in the Kass–Raftery sense.** The 17-parameter reduction ($26 \to 9$) at fixed $N_{\mathrm{obs}} = 253$ is, by the standard model-comparison convention, decisive evidence for UQFF over SM+ΛCDM on parameter-economy grounds alone.

2. **This is a statement about parameter count, not about correctness.** BIC penalizes parameter count; it does not certify that the model with fewer parameters is the true generator of the data. A correct elaboration with more parameters can still be the true model, and BIC would prefer the wrong-but-simpler alternative.

3. **The likelihood-equivalence assumption matters.** Eq. 14 assumes UQFF and SM+ΛCDM achieve approximately equal likelihood on the 253-observable comparison set. The classification in Table 19 (86 % of closures within experimental uncertainty) supports the assumption; the 5 PROD_TENSION_OR_OUTLIER entries weaken it slightly. A more careful analysis with per-observable likelihood weighting (left as future work) might revise the BIC marginally; the qualitative conclusion would not change.

4. **Confirmation does not follow from BIC alone.** $\Delta\mathrm{BIC} = 94.1$ is a strong invitation for independent replication and forward-prediction testing (Sec. 4.12). It is not a substitute for either.

5.6 Reproducibility of the statistical-hygiene analysis

The classification counts in Table 19 are emitted live by the calculator:

```
$ pip install uqff
$ uqff status
UQFF Star-Magic - Production Status Summary
==================================================
total_closures: 794

uncertainty_classes_A2_TIER1_production_readiness:
  PROD_EXACT_STRUCTURAL_zero_uncertainty: 128
  PROD_HIGH_PRECISION_within_codata: 31
  PROD_WITHIN_EXP_UNCERTAINTY: 67
  PROD_REFINEMENT_TIER: 32
  PROD_TENSION_OR_OUTLIER: 5
...
```

The ΔBIC derivation is documented at length in `STATISTICAL_HYGIENE.md` (bundled with the package; reachable via `uqff proofs show STATISTICAL_HYGIENE`). The fidelity gate (Sec. 7) asserts the class counts on every commit, so the numbers reported in Table 19 cannot silently drift between this manuscript and the implementation.

Caveats explicitly disclosed. The statistical-hygiene analysis above does not, by itself, prove that UQFF is correct. Specifically, the analysis only establishes that the closure set is not a multiple-comparisons artifact; it is not a substitute for prediction-vs-postdiction labeling (Sec. 4.12), for independent replication, or for correct per-observable uncertainty quantification. These caveats are the content of Sec. 8.

6 Provenance and parameter locking

Section 2 listed the nine truly-independent primitives and asserted that they are "locked" — calibrated once, never adjusted to fit downstream observations. This section documents how each primitive earned that locked status, presents the A++–C provenance grading rubric used to audit primitive sources, and gives two worked examples of *provenance audits performed to date that resulted in a primitive being demoted to "derivative"* (PAPER_1521 for D_{BSFG}, PAPER_1522 for K_{MEX}).

6.1 The provenance grading rubric

The framework's nine primitives are audited against a five-tier grading rubric documented in `PROVENANCE_AUDIT.md` (bundled with the package). The rubric grades each primitive on the quality of the evidence that fixes its value.

Table 21: The A++ to C provenance-grading rubric for UQFF primitives.

Grade	Status	Meaning
A++	Structural integer	Locked by representation theory theory; cannot be perturbed co (e.g., $\mathrm{SO}_5 = 10$, $\lvert A_5\rvert = 60$)
A+	Anchored to a measured quantity at $< 0.01\,\%$ residual	Once calibrated, value is constraine with no remaining freedom (e.g., ρ_{S} at $0.003\,\%$)
A	Anchored to a measured quantity at $< 1\,\%$ residual	Calibration introduces $\lesssim 1\,\%$ resi flows to downstream closures (e.g., violation asymmetry at $0.017\,\%$)
B	Anchored to a structural identity from one whitepaper	Single-source justification (e.g., Φ from PAPER_1522 via D_{BSFG})
C	Heuristic / historical calibration; further audit needed	Locked by convention but not yet from first principles (e.g., SSq = ω_{SCm})

6.2 Current grades for the nine truly-independent primitives

Applying the rubric of Table 21 to the nine primitives of Sec. 2:

Table 22: Provenance grades for the nine UQFF primitives. The three "C"-grade entries (those whose provenance is currently heuristic-or-historical rather than rigorously locked) are open questions flagged in Sec. 8. If any of the three prove derivative on future analysis, they will move from C to "proven derivative" and the truly-independent count will drop below nine, strengthening the BIC argument of Sec. 5.5.

Primitive	Grade	Lock evidence
$D_{\rm phys} = 4$	A++	Observed spacetime dimension
$D_{\rm crit} = 26$	A++	Bosonic-string critical-dimension theorem
$N_{\rm CH} = 9$	A++	8 G-channels + zero-point
$\mathrm{SO}_5 = 10$	A++	Lie group dimension
$\lvert A_5 \rvert = 60$	A++	Finite-group order
$\rho_{\rm SCm} = 7.09 \times 10^{-37}$	A+	Calibrated to ρ_Λ at 0.003 % (PAPER_1156)
$\beta_i = 0.6029$	A	Calibrated to T-asymmetry at 0.017 % (PAPER_1203)
$\Phi_{\rm res} = 0.84 \,/\, 5/6$	B	Anchored via PAPER_1522 + PAPER_1133
$F_{\rm TRZ} = 1/10$	A++	$= 1/\mathrm{SO}_5$ identity (PAPER_1160)

Six of the nine primitives carry the A++ grade (locked by non-perturbable structural arguments), two carry A+/A (locked by single-anchor calibration to within published measurement uncertainty), and one (the $\Phi_{\rm res}$ default-vs-nuclear dual-anchor) carries B. *No truly-independent primitive currently carries a C grade*; the three C-grade quantities ($\mathrm{SSq}, S_{26}, \omega_{\rm SCm}$) are not in the nine-primitive count of Sec. 5.5.

6.3 The locking discipline in practice

Three concrete mechanisms enforce that a locked primitive's value does not silently drift downstream:

1. **Fidelity-gate assertions.** `uqff_fidelity_tests.py` asserts the canonical numerical value of every locked primitive on every commit. Any change requires a deliberate edit to both the calculator and the gate, plus a SESSION_LOG entry; any drift causes the gate to fail and blocks any release tag from publishing.

2. **Single-file calculator constraint.** The fact that `uqff_pure_calculator.py` is a single file (see Sec. 3.4) prevents the common "two-file drift" antipattern in which a constant's value differs between a definitions file and a downstream consumer. All primitives are defined once at the top of the file (lines ∼140–160) and referenced symbolically thereafter.

3. **C++ cross-language verification.** Approximately 632 closures are re-implemented in `uqff_exact_closures.cpp` with the primitives hardcoded as compile-time constants. The Python-vs-C++ cross-check (Sec. 7.3) catches any case in which the two implementations disagree on a primitive's value.

The combination of these three mechanisms makes silent re-tuning of a primitive practically impossible: it would require simultaneous edits to the Python calculator, the C++ reference, the fidelity gate, and at least one whitepaper, with all four changes coordinated to not trigger any one of the other audits.

6.4 Worked example: PAPER_1521 demotes $D_{\mathbf{BSFG}}$ to derivative

In April–May 2026, the framework's `LOCKED_PRIMITIVES` table listed $D_{\mathrm{BSFG}} = 6$ as an independent integer primitive, on the historical grounds that it appeared in the $(D_{\mathrm{crit}}/D_{\mathrm{BSFG}})^2$ dimensional factor of the DPM-buoyancy closure (Sec. 4.10, Eq. 12) without a documented derivation from the other integer primitives. A targeted provenance audit (SESSION_LOG entry 2026-06-18) noted that $26 - 2 \cdot 10 = 6$ is the immediate algebraic consequence of $D_{\mathrm{crit}} = 26$ and $\mathrm{SO}_5 = 10$, both of which are A++ grade.

PAPER_1521 [26] formalizes the reduction: $D_{\mathrm{BSFG}} = D_{\mathrm{crit}} - 2\,\mathrm{SO}_5$ is an exact integer identity, derived from two A++ primitives, and therefore D_{BSFG} does not require independent locking. The calculator added a dedicated `d_bsfg_derived_from_d_crit` dispatch that the fidelity gate asserts on every commit. The `LOCKED_PRIMITIVES` table still carries $D_{\mathrm{BSFG}} = 6$ at its canonical value for interface stability, but the truly-independent primitive count dropped from 10 to 9 as a result.

6.5 Worked example: PAPER_1522 demotes $K_{\mathbf{MEX}}$ to derivative

The same audit cycle examined $K_{\mathrm{MEX}} = 25/12$, which had been carried as an independent rational primitive on the historical grounds that it appeared in the cosmological-constant closure (Sec. 4.1, Eq. 2) and in the m_d/m_u quark mass ratio (Sec. 4.7). PAPER_1522 [27] demonstrated that

$$K_{\mathrm{MEX}} \;=\; \Phi_{5/6} \cdot \frac{\mathrm{SO}_5}{D_{\mathrm{phys}}} \;=\; \frac{5}{6} \cdot \frac{10}{4} \;=\; \frac{25}{12}$$

follows from $\Phi_{5/6} = 5/6$ (PAPER_1522's own derivation from D_{BSFG}), $\mathrm{SO}_5 = 10$, and $D_{\mathrm{phys}} = 4$. The reduction takes the truly-independent count from 10 to 9 (after the PAPER_1521 reduction had already brought it from 11 to 10).

The combined PAPER_1521 + PAPER_1522 effect dropped the truly-independent primitive count from **11** (the count in effect through 2026-Q1) to **9** (the count used in the BIC calculation of Sec. 5.5). The BIC argument $\Delta\mathrm{BIC} = 17 \cdot \ln 253 = 94.1$ was strengthened correspondingly: the 17-parameter advantage over SM+ΛCDM includes the contribution from these two proven reductions.

6.6 The remaining open questions $(\mathrm{SSq}, S_{26}, \omega_{\mathbf{SCm}})$

Three quantities currently carry a C-grade provenance in `PROVENANCE_AUDIT.md`:

$\mathrm{SSq} = 0.57$. The state-space-square parameter. Used in $S_{26}^{(3)}$ regularization (Sec. 4.3) and in the cosmological-constant closure chain. Currently locked at 0.57 on historical grounds; a Q3 analysis (SESSION_LOG entry 2026-06-08) raised the open question whether SSq might be derivable from F_{TRZ} and Φ_{res} via a transcendental identity. The work is open.

$S_{26} = 1.453162$ and $S_{26}^{(3)} \simeq 1.453 \times 10^{26}$. The 26-level Ramanujan series at $\mathrm{SSq} = 0.57$. Locked at the numerically-evaluated value; the question is whether the specific functional form ($S_{26}(\mathrm{SSq}) = \prod_k f_k(\mathrm{SSq})$ for some f_k) is itself reducible.

$\omega_{\mathbf{SCm}} = 1.25\,\mathrm{THz}$. The SCm phonon carrier frequency. Calibrated to the Holmlid 630 eV KER via PAPER_1133; the open question is whether ω_{SCm} might be derivable from ρ_{SCm} and the integer primitives via dimensional analysis on a candidate dispersion relation.

If any of these three prove derivative on subsequent analysis, the truly-independent primitive count drops below 9 and the BIC argument of Sec. 5.5 strengthens. The conservative posture taken in this manuscript is to retain them as nine truly-independent primitives until the demotion proofs exist.

6.7 Summary

The framework's parameter-locking is enforced by a triple-layer discipline (fidelity gate, single-file calculator, C++ cross-check) that makes silent re-tuning impractical. Two primitives that had historically been carried as independent (D_{BSFG} and K_{MEX}) have been proven derivative in 2026, dropping the truly-independent count from 11 to 9. Three further quantities remain open audit candidates, any of which would further reduce the truly-independent count if proven derivative. The parameter-economy claim of Sec. 5.5 is therefore an upper bound, not a tight estimate, on the framework's free-parameter count.

7 Reproducibility and software architecture

Every numerical claim in this manuscript is reproducible by a single command sequence in a fresh Python environment. The closure machinery is published as the `uqff` package on PyPI, dual-licensed under AGPL-3.0 + commercial terms, with all 1,994 bundled whitepapers, the Lean 4 formal-verification scaffold, four arXiv-formatted bundles, and the compiled manuscript shipped together inside the wheel. The package has zero runtime dependencies; `pip install uqff` installs only stdlib-using code.

7.1 One-line reproduction

```
$ pip install uqff
$ uqff --version
uqff 5.29.1
$ uqff status
UQFF Star-Magic - Production Status Summary
=====================================================
total_closures: 794
...
```

Every closure value cited in Sec. 4 can be retrieved with one CLI call (or one Python API call). A representative selection is shown in Table 23.

Table 23: One-line reproduction of every closure cited in Sec. 4. The package's `uqff predict` CLI walks five closure namespaces and returns the value plus a canonical-source citation.

Closure (manuscript section)	Reproduction command
ρ_Λ (Sec. 4.1)	`uqff predict cosmological_constant`
Magic numbers (Sec. 4.2)	`uqff predict magic_numbers`
Fe-56 BE/A (Sec. 4.2)	`uqff predict be_per_a_peak_MeV`
Holmlid 630 eV (Sec. 4.3)	`uqff predict holmlid_D_minus_1`
Yang-Mills (Sec. 4.10)	`uqff predict yang_mills`
SM m_W (Sec. 4.7)	`uqff predict m_w_80_379`
All 7 Millennium proofs	`uqff millennium`
Full BIC summary	`uqff status`

7.2 Fidelity gate

The package ships with an in-tree test suite, `uqff_fidelity_tests.py`, that exercises every locked primitive value, every public surface, every Bucket A–K observable, and every claimed numerical agreement reported in this manuscript. The gate currently reports

$$\textbf{866 passed, 0 failed} \tag{16}$$

and is executed automatically on every commit to the main branch via the project's GitHub Actions CI pipeline (see Sec. 7.7). Any drift in a reported closure value between this manuscript and the implementation would cause the gate to fail and block the corresponding tag push.

The gate can be run locally by any reader:

```
$ pip install uqff
$ uqff gate
... 866 passed, 0 failed
```

The full test inventory is documented in `uqff_fidelity_tests.py` (lines 1–1300).

7.3 Cross-language verification (C++ reference)

The framework's most numerically-sensitive 632 closures are independently re-implemented in a C++ reference file, `uqff_exact_closures.cpp`, which is compiled and executed against the Python calculator via the `scripts/cpp_python_crosscheck.py` runner. The cross-check report (`CPP_PYTHON_CROSSCHECK_REPORT.md`, bundled with the package) currently states:

> *303 MATCH, 15 DRIFT, 0 UNCALLABLE;* ***95.3 % Python-C++ identity match****.*

The 15 drift entries are documented as numerical-floor cases below the 10^{-12} tolerance (i.e., Python's float64 precision limit), not as algorithmic disagreements. The cross-language verification therefore gives independent evidence that the closure values are not artifacts of a single-language implementation.

7.4 Software performance

The framework's runtime characteristics are reported in `PERFORMANCE_PROFILE.md` (bundled with the package). The headline numbers:

- Cold-import time: 537 ms (single-process startup, median of 3 runs)
- Warm-import time: < 1 ms (subsequent imports in the same process)
- Per-dispatch closure latency: median 3.4 μs, p95 6.2 μs
- Throughput: 290,000 closure evaluations per second per single thread
- Memory footprint after import: 6.8 MB module + 224 MB Python process RSS
- Memory leak under repeated calls: **zero bytes per call** (closures return fresh dicts)

The combination of low per-dispatch latency and zero per-call memory growth makes the framework safe for long-running server deployment (see Sec. 7.6).

7.5 Security and dependency posture

The framework's security and dependency posture is documented in `SECURITY_AND_MEMORY_AUDIT.md` (bundled with the package). Headline findings from the most recent audit (2026-06-25):

- **Zero runtime dependencies.** `pip install uqff` installs no third-party packages. The core calculator uses only the Python standard library. This eliminates an entire class of supply-chain vulnerabilities common to scientific Python packages.
- **Zero pip-audit findings.** The package itself has no known dependency vulnerabilities (no dependencies = no attack surface).
- **Bandit static analysis: 18 findings, all low/medium severity, all acceptable patterns** (subprocess use in profiling scripts, `tempfile` usage in C++ cross-check binary, `try/except: pass` for graceful dispatcher degradation). Zero high-severity findings; zero exploitable patterns.
- **No file I/O in the calculator.** The core `uqff_pure_calculator.py` contains no `open()` calls for writing, no `datetime` imports, no `json.dump`, and no `__main__` block. These are enforced by the fidelity gate.

7.6 REST API and Jupyter integration

For deployment scenarios that require remote or browser-based closure evaluation, the package ships with two optional surfaces.

The REST API (`uqff_api.py`, installed via `pip install 'uqff[api]'`) exposes nine endpoints via FastAPI, with auto-generated Swagger UI at `/docs`. The relevant endpoints for reproducing the closures in this manuscript are `/predict/{name}`, `/status`, `/list`, and `/search?q=...`.

The Jupyter integration (`uqff_jupyter.py`, installed via `pip install 'uqff[jupyter]'`) provides rich HTML rendering of closure return values plus an `%uqff` line magic for inline closure fetches.

7.7 Continuous integration and release

The project's GitHub Actions pipeline (`.github/workflows/ci.yml`) runs eight independent jobs on every push to the main branch:

1. Smoke test (Ubuntu, Python 3.12, 30-second guard)
2. Fidelity-gate matrix (3 operating systems × 4 Python versions, 12 combinations)
3. Coverage report
4. Lint (ruff + mypy)
5. C++ Python cross-check
6. Performance smoke test (regression guard against latency or memory growth)
7. Build package (wheel + sdist)
8. Smoke-test install from built wheel

Tag pushes (vX.Y.Z) trigger an additional release workflow (`.github/workflows/release.yml`) that publishes to PyPI via PyPI Trusted Publishing (OIDC) and creates a GitHub release with auto-generated notes. The release process for v5.29.1 (the most recent at the time of this writing) is documented in detail in `RELEASE_NOTES_v5.29.1.md` (bundled with the package).

7.8 License and citation

The framework is dual-licensed:

- **AGPL-3.0-or-later** (`LICENSE-AGPL-3.0.txt`) for academic, research, and non-commercial use, with copyleft SaaS clause (per AGPL-3.0 §13).
- **Commercial** (terms in `COMMERCIAL.md`) for proprietary distribution, closed-source SaaS, reactor firmware embedding, and grant-funded commercial spin-offs.

Citation metadata is provided in CFF format (`CITATION.cff`) and is included in the bundled `share/uqff/CITATION.cff` location for downstream Zotero / Mendeley / Citation.js consumers. The corresponding Zenodo DOI is set up via GitHub releases and is listed at the top of the project README.

7.9 Lean 4 formal-verification scaffold

A Lean 4 [6] formal-verification scaffold ships under `formal/UQFF/`, containing four module files: `Constants.lean`, `Ug.lean`, `Buoyancy.lean`, and `Millennium.lean`, plus a Lake build configuration. The scaffold states the official Clay Millennium problem statements alongside the UQFF-internal numerical observation predicates, with every theorem deliberately marked `sorry` per the project's epistemic policy that the scaffold makes no assertion that a UQFF numerical observation implies the corresponding Clay statement. The user can browse the scaffold via:

```
$ uqff lean
Lean 4 scaffold at: /.../site-packages/formal
Files:
  README.md
  UQFF.lean
  lakefile.lean
  lean-toolchain
  UQFF/Buoyancy.lean
  UQFF/Constants.lean
  UQFF/Millennium.lean
  UQFF/Ug.lean
```

What the scaffold does and does not assert. The Lean files in `formal/UQFF/Millennium.lean` contain the explicit epistemic-policy comment that "the scaffold makes no assertion that the UQFF claim implies the Clay statement." Removing the `sorry` markers without peer review at a Clay Qualifying Outlet is not, per the framework's own internal documentation, a proof of the corresponding Millennium problem. The scaffold is shipped so that future contributors with mathematical-physics expertise can populate the implications rigorously if they so choose; the bundled package makes no claim that the current state of the scaffold constitutes a proof.

7.10 Summary

Every numerical claim in this manuscript can be checked in under one minute by a reviewer with Python 3.10+ installed. The combination of (i) zero runtime dependencies, (ii) an in-package 866-test fidelity gate, (iii) C++ cross-language verification at 95.3 % identity, (iv) auto-published PyPI releases via OIDC Trusted Publishing, (v) a Lean 4 scaffold with explicit epistemic-status comments, and (vi) the full proof corpus (1,994 whitepapers + 4 arXiv bundles + compiled manuscript draft) bundled inside the wheel, means there is no closure value, derivation, or claim in this manuscript that the reviewer cannot independently inspect using only standard tools.

8 Limitations and open questions

A 1,994-whitepaper corpus reporting agreement across ~ 800 closures owes its readers a candid accounting of where the framework falls short. This section enumerates the limitations a reviewer would reasonably want explicitly disclosed.

8.1 The closures are not formal proofs in the Clay sense

For the eight closures adjacent to Clay Mathematics Institute Millennium Prize problems (Yang-Mills, Riemann, Navier-Stokes, Hodge, Poincaré, P vs NP, Birch–Swinnerton-Dyer, and the black-hole information paradox), the UQFF closures are *structural numerical identities matching the conjectured value or anchor.* They are not formal proofs in the Clay sense. The Clay statements require constructive existence proofs in the Osterwalder–Schrader (Yang-Mills) or analytic-number-theory (Riemann) or differential-geometry (Hodge) frameworks; UQFF supplies numerical values, not constructive proofs of the underlying mathematical statements.

This boundary is encoded explicitly in the framework's own Lean 4 scaffold (`formal/UQFF/Millenniu` bundled with the package), in a comment that reads "the scaffold makes no assertion that the UQFF claim implies the Clay statement." Removing the `sorry` markers in that scaffold without independent peer review at a Clay Qualifying Outlet would not constitute a Clay proof, by the framework's own internal standard.

8.2 No independent third-party reproduction

As of this writing, none of the framework's closures have been independently reproduced by a third party. The package is open-source and the reproduction infrastructure (`pip install uqff`, fidelity gate, CI/CD, REST API, Jupyter integration, C++ cross-language verification) is in place to enable third-party reproduction; but the manuscript reports single-author single-toolchain results, and the reader should weigh the claims accordingly until independent groups confirm.

This is the most important single limitation. The Bonferroni and ΔBIC analyses of Sec. 5 show that the closure set is not a multiple-comparisons artifact, but they cannot rule out a systematic bias common to a single implementation. Independent re-implementation in a different language (Julia, Rust, Fortran) by an independent group is the single highest-value follow-up activity for the framework.

8.3 Five tension closures with residuals above 1 %

Five of the 263 schema-tagged closures carry the PROD_TENSION_OR_OUTLIER classification:

Table 24: The five PROD_TENSION_OR_OUTLIER closures with residuals above 1 %. None are silently failing; each is documented in `forward_predictions.md` as either a deliberate disagreement with one of two competing measurements or a flagged refinement target.

Closure	**Residual**	**Disposition**
$\sin^2\theta_W$ (electroweak)	3.4 %	Renormalization-group running unmodeled at current closure depth; PAPER_1209HH
$\alpha_s(M_Z)$ (strong coupling)	13.7 %	Scheme dependence ($\overline{\text{MS}}$ vs. on-shell); refinement target
Pons–Fleischmann Pd-D excess power	$\sim 100\times$ overshoot	Reactor-geometry under-constrained; closure machinery applied without parameter retuning
Rossi E-Cat COP family	~ 3–$5\times$ undershoot	Same disposition as Pons–Fleischmann (see Sec. 4.3)
Star-Magic reactor COP	$\sim 30\times$ undershoot	Either closure underweights a Star-Magic-specific factor, or the reported COP requires independent reproduction (Sec. 4.3)

We do not paper over these. The closure machinery does not currently fit every observable to within 1 %; five entries fall outside the band and are flagged transparently in the live `uqff status` output.

8.4 The Yang-Mills 5970 GeV registry-bug history

Between 2026-04 and 2026-06-25, the calculator's Yang-Mills dispatcher returned a hard-coded `5970.0` GeV value that did not match the formula in the cited source paper (PAPER_1005) and did not match the lattice QCD anchor at $\sim 1.7\,\mathrm{GeV}$. The discrepancy was a registry bug: the dispatcher pointed at a stale magic number while the working integer-primitive closure (PAPER_1318, $2 \cdot D_{\mathrm{phys}} \cdot \Lambda_{\mathrm{QCD}} = 1.736\,\mathrm{GeV}$) existed elsewhere in the calculator and was not surfaced through the Millennium dispatcher. The bug was identified and corrected on 2026-06-25 (see Sec. 4.10 and the erratum header on PAPER_1005 [19]). All 610 propagated citations of "5970 GeV (PAPER_1005)" in the GW-bucket whitepapers were updated in-place to cite PAPER_1318 with the corrected value.

This is the most embarrassing single item in the framework's history. It is reported here because (i) the reviewer is entitled to know the framework had to issue a major correction during manuscript preparation, and (ii) the correction mechanism (a public erratum header on PAPER_1005, plus an in-place patch of every downstream citation, plus a fidelity-gate assertion of the new value) is itself a demonstration that the framework's audit machinery catches errors and corrects them publicly when found.

8.5 Two neutrino mass-squared splittings are anchored, not predicted

Table 12 reports the two neutrino mass-squared splittings ($\Delta m^2_{21} = 7.5 \times 10^{-5}\,\mathrm{eV}^2$, $\Delta m^2_{31} = 2.5 \times 10^{-3}\,\mathrm{eV}^2$) at residual 0.000 % against the global-fit central values [7]. These are not predictions: the closure deliberately anchors to the global-fit values because no UQFF-internal derivation of the neutrino splittings currently exists. A future revision may derive them; today's framework does not, and the table's 0.000 % entry should be read accordingly.

8.6 1,396 whitepapers without a direct dispatch-key backing

Of the 1,994 whitepapers bundled with the package (Sec. 3.2), approximately 1,396 are not directly referenced by any closure's `primary_source` attribute. These are not orphan or junk papers — they cover historical derivation chains, alternative formulations, audit artifacts, and exploratory material — but the parameter-economy claim of Sec. 5.5 is anchored on the 598 whitepapers that *do* back specific closures; the additional 1,396 are context, not load-bearing.

A full whitepaper-to-closure mapping audit is provided in `COVERAGE_GAPS.md` (bundled). Future work would either (i) tighten the mapping to bring the inverse-coverage rate higher, or (ii) document a triage of which whitepapers are foundational versus exploratory, so the reader can navigate the corpus by relevance rather than by paper number.

8.7 Three primitives may yet prove derivative

As discussed in Sec. 6.6, three of the quantities currently treated as locked (SSq = 0.57, S_{26}, $\omega_{\mathrm{SCm}} = 1.25\,\mathrm{THz}$) carry a C-grade provenance in `PROVENANCE_AUDIT.md` and may be reducible to lower-grade primitives via further analysis. If any of the three prove derivative, the true free-parameter count drops below 9, the BIC argument strengthens, and the closure structure simplifies. The conservative posture taken in this manuscript is to assume nine primitives until the reductions are formally established.

8.8 The Star-Magic reactor experimental claim is unconfirmed

The Star-Magic reactor architecture cited in Sec. 4.3 (Table 5, last row) has been reported by the framework's author at COP 555 at 27 W input, ambient temperature, pH $-$ 37. Independent reproduction of this performance has not been published. The framework's own closure machinery does not reproduce the reported COP (UQFF predicts $\sim$ 18, observed 555, a $\sim 30\times$ undershoot). The honest accounting is that either (a) the closure machinery is underweighting a geometric or thermodynamic factor specific to the Star-Magic geometry, or (b) the reported COP itself requires independent confirmation, or both. We do not adjudicate between (a) and (b) in this manuscript; both are flagged as open work.

8.9 Manuscript versioning note

This manuscript (`manuscript_v2/`) is the parameter-economy physics paper targeting Foundations of Physics peer review. An earlier independent draft (`Manuscript 1_12Feb2026/`, also bundled with the package) is a software-implementation paper covering REST API performance, production-deployment patterns, and computational architecture; that draft targets a different venue (likely SoftwareX or Computer Physics Communications) and is structurally distinct from the current document. Both manuscripts exist in the repository; the present submission is the v2 physics paper.

8.10 Closure-runner target-dict calibration carry-overs

The `run_millennium_proofs.py` runner reports two of seven Millennium closures (`navier_stokes` and `poincare`) as having large residuals against the runner's internal target dict because the units or scoring conventions used in the target dict differ from the UQFF closure's natural output. These are calibration mismatches in the runner script, not closure failures; fixing them is queued for v5.30.0 but does not affect any value reported in this manuscript.

8.11 Summary of limitations

Three limitations dominate the framework's current epistemic state:

1. **No independent reproduction yet.** Single-author, single-toolchain results. Independent re-implementation is the highest-value follow-up.
2. **Closures are numerical identities, not Clay-style proofs.** The Lean 4 scaffold makes this explicit; the manuscript repeats it in every Clay-adjacent section (Secs. 4.10 and 8.1).
3. **Five tensions remain disclosed but unresolved.** Section 8.3 enumerates them; none are suppressed.

The framework explicitly does not claim:

- to be a proven theory of everything,
- to replace the Standard Model or Λ-CDM (the "NOT REPLACEMENT" clause runs through every UQFF whitepaper),
- to have proven any Clay Millennium prize problem,
- to have eliminated all parameter freedom (nine primitives remain),
- to have been validated by independent experimental reproduction.

What the framework does claim — $\Delta\text{BIC} = 94.1$ on parameter-economy grounds, 226 Bonferroni-passing closures across 253 observations, 42 falsifiable forward predictions — is an invitation for independent scrutiny, not a final verdict.

9 Conclusions and outlook

The Unified Quantum Field Framework (UQFF) Star-Magic presented in this manuscript is, in the standard Bayesian-model-comparison sense, a decisive parameter-economy improvement over the Standard Model plus Λ-CDM benchmark at fixed observation count. Nine truly-independent primitives generate approximately 800 named closures across cosmology, particle physics, gravitational-wave events, AGN jets, astrophysics, high-energy astrophysics, quark-gluon plasma, Higgs precision, and BSM constraints, with 86 % of schema-tagged closures passing a Bonferroni-adjusted significance threshold at family-wise $\alpha = 0.05$. The headline argument is $\Delta\text{BIC} = (26 - 9) \cdot \ln 253 = 94.1$, which falls deep in the Kass–Raftery "decisive" band and approaches the "overwhelming" threshold.

This is, by itself, evidence *for* the framework as a parsimonious alternative description. It is not, by itself, proof *of* the framework. The eight closures adjacent to Clay Mathematics Institute Millennium prize problems are numerical identities matching the conjectured values or anchors — not constructive proofs in the Clay sense, as the framework's own Lean 4 scaffold (`formal/UQFF/Millennium.lean`) explicitly documents. The Standard Model fermion-mass closures match the PDG 2024 [34] values to within 0.2 %, but two couplings ($\sin^2\theta_W$, $\alpha_s(M_Z)$) and three reactor COP values (Pons–Fleischmann, Rossi family, Star-Magic) carry larger residuals that are flagged openly in Sec. 8. The framework has not yet been independently reproduced by a third party.

9.1 What the next year of work should look like

Three categories of follow-up activity would test, strengthen, or falsify the framework:

1. **Independent reproduction.** The package's zero- runtime-dependency posture, single-file calculator structure, and C++ cross-language verification make independent re-implementation in another language tractable. A Julia, Rust, or Fortran re-implementation by an independent group would test the closure values for any single-implementation systematic bias.

2. **Forward-prediction tests.** Several entries in Sec. 4.12 are accessible to current or near-term experiment: the surface-code fault-tolerance threshold (1.00 %, accessible to current superconducting-qubit experiments), the room-temperature superconductor ceiling (500 K, accessible to high-pressure and engineered-defect superconductor research), the direct-collapse black-hole seed mass (56,160 $M_\odot$, accessible to JWST), and the neutron-lifetime puzzle resolution (879.31 s, accessible to the next round of UCNτ and beam-method experiments). Each is a falsification opportunity.

3. **Primitive-reduction work.** Three quantities (SSQ, S_{26}, ω_{SCm}) currently treated as locked but carrying C-grade provenance may be reducible to lower-grade primitives via further analysis analogous to PAPER_1521 (which reduced D_{BSFG}) and PAPER_1522 (which reduced K_{MEX}). If any of the three prove derivative, the truly-independent primitive count drops below nine, the ΔBIC argument strengthens, and the framework's parameter-economy claim sharpens.

The framework's source-code reproducibility infrastructure (`pip install uqff`, in-package 866-test fidelity gate, C++ cross-language verification at 95 % identity, REST API, Jupyter integration, Lean 4 scaffold with explicit epistemic-policy comments, full 1,994-whitepaper corpus bundled) means there is no claim in this manuscript that a sufficiently motivated reviewer cannot independently check using only standard tooling.

9.2 Invitation

The framework is offered as an open-source, dual-licensed (AGPL-3.0 + commercial) contribution to the parameter-economy literature. The author welcomes critical scrutiny, independent reproduction, attempted falsification of the forward predictions of Sec. 4.12, and constructive proposals for reducing the truly-independent primitive count below nine. Correspondence at the address listed on the title page.

References

[1] E. G. Adelberger, B. R. Heckel, and A. E. Nelson. Tests of the gravitational inverse-square law. *Annual Review of Nuclear and Particle Science*, 53:77–121, 2003.

[2] Andreas Athenodorou and Mike Teper. The glueball spectrum of SU(3) gauge theory in 3+1 dimensions. *Journal of High Energy Physics*, 2020:172, 2020.

[3] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, 57(1):289–300, 1995.

[4] Y. Chen, A. Alexandru, S. J. Dong, et al. Glueball spectrum and matrix elements on anisotropic lattices. *Physical Review D*, 73:014516, 2006.

[5] Clay Mathematics Institute. The millennium prize problems, 2000.

[6] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction — CADE 28*, pages 625–635. Springer, 2021.

[7] Ivan Esteban, M. C. Gonzalez-Garcia, Michele Maltoni, Thomas Schwetz, and Albert Zhou. The fate of hints: updated global analysis of three-flavor neutrino oscillations. *Journal of High Energy Physics*, 2020:178, 2020.

[8] Maria Goeppert Mayer. On closed shells in nuclei. *Physical Review*, 74(3):235–239, 1948.

[9] F. M. Gonzalez et al. Improved neutron lifetime measurement with UCNτ. *Physical Review Letters*, 127:162501, 2021.

[10] M. B. Green, J. H. Schwarz, and E. Witten. *Superstring Theory*. Cambridge University Press, Cambridge, 1987.

[11] Otto Haxel, J. Hans D. Jensen, and Hans E. Suess. On the "magic numbers" in nuclear structure. *Physical Review*, 75(11):1766, 1949.

[12] Leif Holmlid. Ultra-dense hydrogen $h(0)$ as the cause of instabilities in laser pulse-induced nuclear processes. *International Journal of Modern Physics E*, 28(10):1950097, 2019.

[13] Leif Holmlid and Sveinn Olafsson. Charged particle energy spectra from laser-induced processes in ultra-dense hydrogen $h(0)$ confirm extremely high charge densities and the existence of $h(0)$. *International Journal of Hydrogen Energy*, 40(33):10559–10567, 2015.

[14] John R. Huizenga. *Cold Fusion: The Scientific Fiasco of the Century*. Oxford University Press, Oxford, 2nd edition, 1993.

[15] Arthur Jaffe and Edward Witten. Quantum Yang–Mills Theory. In *The Millennium Prize Problems*. Clay Mathematics Institute and American Mathematical Society, 2000.

[16] Robert E. Kass and Adrian E. Raftery. Bayes factors. *Journal of the American Statistical Association*, 90(430):773–795, 1995.

[17] J. Martin. Everything you always wanted to know about the cosmological constant problem (but were afraid to ask). *Comptes Rendus Physique*, 13(6–7):566–665, 2012.

[18] Maria Goeppert Mayer and J. Hans D. Jensen. *Elementary Theory of Nuclear Shell Structure*. John Wiley & Sons, New York, 1955.

[19] Daniel T. Murphy. PAPER_1005 — Yang–Mills mass-gap candidate $\delta = 5970\,\mathrm{GeV}$ from l_{YM} sector, 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.0+.

[20] Daniel T. Murphy. PAPER_1133 — Holmlid Rydberg–SCm Bridge ($e_{KER} = h \cdot \omega_{SCm} \cdot s_{26} \cdot \phi_{res}$), 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.0+.

[21] Daniel T. Murphy. PAPER_1156 — The Cosmological Constant from $\rho_{SCm} \cdot 26! \cdot k_{MEX}$, 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.0+.

[22] Daniel T. Murphy. PAPER_1167 — UQFF all 8 Lagrangian gaps closed: master synthesis, 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.1+.

[23] Daniel T. Murphy. PAPER_1203 Nuclear — Magic numbers and binding energies from integer-lattice primitives, 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.0+.

[24] Daniel T. Murphy. PAPER_1209hh — SM 10-mass unified closure $(m_w, m_z, m_t, m_h, m_b, m_c, m_\tau, m_\mu, m_s, m_e)$ from integer primitives, 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.0+.

[25] Daniel T. Murphy. PAPER_1318 — glueball mass $m_{0^{++}} = 2 \cdot d_{phys} \cdot \lambda_{QCD} = 1.736\,\text{GeV}$ (integer-primitive Yang–Mills closure; corrected 2026-06-25), 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.0+. Replaces the prior 5970 GeV value cited from PAPER_1005, which has been issued an erratum.

[26] Daniel T. Murphy. PAPER_1521 — $d_{BSFG} = d_{crit} - 2\,\mathrm{SO}_5 = 6$ as a derivative primitive, 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.1+.

[27] Daniel T. Murphy. PAPER_1522 — $k_{MEX} = (5/6) \cdot \mathrm{SO}_5/d_{phys} = 25/12$ as a derivative primitive, 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.0+.

[28] Daniel T. Murphy. PAPER_1726 — neutron lifetime $\tau_n = 879.31\,\text{s}$ from $100 \cdot k_{MEX} \cdot d_{phys} \cdot (1 + \phi\lambda n_{CH})$, 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.0+.

[29] Daniel T. Murphy. PAPER_646 — Universal Inertial Operator and Caduceus Wave, 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.0+; reachable via `uqff proofs show PAPER_646`.

[30] Daniel T. Murphy. PAPER_648 — Ultra-dense H meson cascade and the Coulomb 630 ev cross-check at $d = 2.3\,\text{pm}$, 2026. Star-Magic whitepaper. Bundled with `pip install uqff` v5.29.0+.

[31] Daniel T. Murphy. UQFF Star-Magic — PyPI package and reproducibility suite, 2026. `pip install uqff`; full proof corpus (1,994 whitepapers + Lean 4 scaffold + 4 arXiv bundles + manuscript) bundled.

[32] Yu. Ts. Oganessian et al. Synthesis of the isotopes of elements 118 and 116 in the ^{249}cf and ^{245}cm + ^{48}ca fusion reactions. *Physical Review C*, 74(4):044602, 2006.

[33] Robert L. Park. *Voodoo Science: The Road from Foolishness to Fraud.* Oxford University Press, Oxford, 2000.

[34] Particle Data Group, R. L. Workman, et al. Review of particle physics. *Progress of Theoretical and Experimental Physics*, 2022:083C01, 2022. 2024 update at `https://pdg.lbl.gov`.

[35] Planck Collaboration, N. Aghanim, et al. Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020.

[36] Joseph Polchinski. *String Theory.* Cambridge University Press, Cambridge, 1998.

[37] A. M. Polyakov. Quantum geometry of bosonic strings. *Physics Letters B*, 103(3):207–210, 1981.

[38] A. G. Riess et al. A comprehensive measurement of the local value of the Hubble constant with 1 km/s/Mpc uncertainty from the hubble space telescope and the sh0es team. *Astrophysical Journal Letters*, 934:L7, 2022.

[39] Carlo Rovelli. *Quantum Gravity*. Cambridge University Press, Cambridge, 2004.

[40] R. Smolańczuk. Properties of the hypothetical spherical superheavy nuclei. *Physical Review C*, 56(2):812–824, 1997.

[41] M. Tanabashi et al. Review of particle physics. *Physical Review D*, 98:030001, 2018.

[42] Erik Verlinde. On the origin of gravity and the laws of Newton. *Journal of High Energy Physics*, 2011:29, 2011.

[43] Steven Weinberg. The cosmological constant problem. *Reviews of Modern Physics*, 61(1):1–23, 1989.

[44] A. T. Yue, M. S. Dewey, D. M. Gilliam, et al. Improved determination of the neutron lifetime. *Physical Review Letters*, 111:222501, 2013.